

Data Analytics Capstone Topic

Final Report

**Multi-Horizon Forecasting of Solar and Wind
Energy Using Hybrid Deep Learning Models**

Walsh College

QM640 V1: Data Analytics Capstone

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GITHUB LINK

Code:- https://github.com/raghav0410/Capstone_QM640-V1-Data-Analytics-Capstone.git

Data:- https://github.com/raghav0410/Capstone_QM640-V1-Data-Analytics-Capstone-Data.git

Abstract

The increasing penetration of renewable energy sources (RES) such as solar and wind has created significant challenges for modern power systems due to their intermittency and variability. Accurate forecasting across multiple time horizons is crucial for grid stability, energy trading, and integration of RES into energy markets. This study investigates multi-horizon forecasting (15-minute, 1-hour, and 24-hour) of solar and wind energy generation using a combination of classical statistical models (Persistence, Seasonal Naïve, ARIMA/SARIMA, XGBoost) and advanced hybrid deep learning models (CNN-LSTM, Temporal Convolutional Networks (TCN), Temporal Fusion Transformers (TFT)).

The analysis leverages global, open-access datasets including NSRDB, ERA5, OPSD, Global Solar Atlas, Global Wind Atlas, Renewables.ninja, and Global Renewables Watch. Data preprocessing involved filtering daytime-only values, removing calm wind periods, handling missing values, and performing extensive Exploratory Data Analysis (EDA) to understand seasonal, diurnal, and geographical variations.

The study addresses four research questions:

- (1) The effect of time horizon on forecast accuracy,
- (2) Performance comparison between hybrid DL models and traditional methods,
- (3) Identification of the most influential meteorological drivers, and
- (4) The transferability of models across different regions. Statistical validation was performed using Wilcoxon signed-rank tests, Diebold–Mariano tests, SHAP analysis, ablation experiments, and TOST non-inferiority tests.

Key findings include:

- (i) Forecast accuracy declines with increasing horizon, especially from 1h to 24h;
- (ii) Hybrid DL models outperform classical baselines, with CNN-LSTM improving solar forecasts by ~15% and TCN improving wind forecasts by ~6–7% over the best baselines;
- (iii) Solar forecasts are primarily driven by irradiance variables (GHI, DNI, DHI), while wind forecasts rely on 100m wind speed and air density; and
- (iv) Pooled models are non-inferior to regional models at short horizons but underperform at 24h, though transfer learning benefits data-scarce regions.

This research provides a reproducible forecasting framework that demonstrates the value of hybrid DL models for renewable energy forecasting and outlines future directions involving Bayesian deep learning for uncertainty quantification. The findings can inform grid operators, policymakers, and renewable energy developers in enhancing energy management and accelerating the global transition to sustainable energy.

1. Introduction

1.1 Background and Motivation

Global energy systems are undergoing rapid transformation, with renewable energy sources (RES) such as solar and wind playing a central role in achieving decarbonization goals. However, their variability due to weather patterns and diurnal cycles introduces significant challenges in maintaining grid stability and efficient energy market operations. Accurate forecasting of renewable generation across multiple horizons (from minutes to days ahead) is therefore vital for grid balancing, storage scheduling, and economic dispatch.

1.2 Problem Statement

While short-term forecasting methods ($\leq 6h$) are well studied, multi-horizon forecasting (15m, 1h, 24h) across global sites remains underexplored, particularly in data-scarce regions. Existing statistical and machine learning models often fail to capture the complex non-linear temporal and spatial dependencies inherent in RES data.

1.3 Research Questions

The study addresses the following research questions:

- RQ1: How does forecast accuracy change across different horizons (15m, 1h, 24h)?
- RQ2: Do hybrid deep learning models outperform traditional statistical baselines?
- RQ3: Which meteorological drivers are most influential for solar and wind forecasting?
- RQ4: Can models trained in one region generalize effectively across regions (transfer learning and pooled models)?

1.4 Significance

By developing a reproducible hybrid DL forecasting framework, this project contributes to the literature on renewable energy forecasting while providing practical tools for grid operators, energy traders, and policymakers. The findings address the dual need for accuracy and transferability in renewable energy forecasting at scale.

2. Literature Review

Forecasting renewable energy has become a critical research area as solar and wind penetration increases worldwide. Traditional statistical models such as ARIMA and SARIMA are widely used but struggle with the nonlinear and highly variable behaviour of renewable resources. This limitation has led to the adoption of machine learning (ML) and deep learning (DL) methods, which better capture nonlinear dynamics and temporal dependencies in large-scale datasets (Benti et al., 2023).

Mishra and Palanisamy (2018) developed a recurrent neural network (RNN) framework capable of producing forecasts across multiple horizons within a single architecture, eliminating the need for separate models for each horizon. Building on this, Simeunović et al. (2021) applied spatio-temporal graph neural networks (ST-GNNs) to photovoltaic (PV) forecasting and showed that incorporating spatial dependencies among sites improves

prediction accuracy. Both studies highlight that sequence-based models can capture horizon-wide patterns and regional interactions effectively.

In wind energy research, Huang and Deng (2023) proposed a spatiotemporal deep neural network that integrates numerical weather prediction (NWP) features with sequence learning, achieving higher accuracy for fine-grained multi-horizon forecasts. Similarly, Roseline et al. (2022) demonstrated that neural networks modeling hybrid renewable systems could effectively represent interactions between solar and wind resources, emphasizing the value of combined approaches.

Data availability is another crucial factor for reproducibility and transferability. Effenberger and Ludwig (2022) reviewed open-source wind datasets and found that accuracy depends heavily on the underlying dataset, including terrain complexity and spatial resolution. At a global level, Robinson et al. (2025) introduced the *Global Renewables Watch* dataset, which applies deep learning to satellite imagery to map solar and wind installations worldwide. These works illustrate the importance of both open datasets and scalable modeling frameworks.

From a methodological perspective, Chen and Guestrin (2016) introduced XGBoost, a tree-boosting algorithm that has become a dominant benchmark for tabular prediction tasks due to its speed, scalability, and accuracy. In the deep learning domain, Oreshkin et al. (2020) proposed N-BEATS, a neural architecture designed for interpretable time-series forecasting, achieving strong performance without handcrafted domain features. Lim et al. (2021) extended sequence forecasting further with the Temporal Fusion Transformer (TFT), which combines recurrent layers with attention mechanisms and variable selection to enhance both accuracy and interpretability.

Uncertainty quantification is increasingly emphasized in renewable forecasting. A Bayesian BiLSTM framework (2022) showed that probabilistic forecasts with calibrated intervals are more informative for operators than single point estimates, especially for day-ahead horizons. To ensure that improvements are statistically valid, Whittle (2025) advanced sample size and power methodologies tailored to prediction models, enabling robust evaluation of performance differences such as reductions in mean absolute error (MAE).

Taken together, these studies underscore four key themes: (1) sequence and attention-based models improve accuracy across horizons; (2) spatiotemporal and hybrid renewable modeling enhance performance by exploiting spatial and cross-resource dependencies; (3) dataset quality and open-source availability are essential for reproducibility and transferability; and (4) uncertainty quantification and statistical rigor are critical for operational relevance. These themes directly align with the present research, which investigates horizon-specific accuracy, hybrid DL performance, feature importance, and transferability of models across regions.

3. Materials and Method

3.1 Data Sources and Aggregation

The foundation of this study lies in the creation of a unified, high-resolution dataset that integrates multiple open-access sources of renewable energy and meteorological data. Forecasting accuracy depends heavily on the quality, consistency, and temporal granularity of the input data; therefore, careful selection and preprocessing were necessary.

The National Solar Radiation Database (NSRDB) was chosen as the primary source for U.S.-based solar irradiance and meteorological variables. It offers high-resolution time-series data, making it particularly well suited for training short-horizon forecasting models. For global meteorological variables, we used the ERA5 reanalysis dataset, which provides hourly data on wind speeds at different heights (10 m and 100 m), air temperature, pressure, and density. ERA5 was indispensable because it provides globally consistent atmospheric coverage, allowing the study to extend beyond regions with dense ground measurements.

Additional data were sourced from the Open Power System Data (OPSD) platform, which provides information on European grid generation and demand. This was complemented by the Global Solar and Wind Atlas, which contains long-term information on renewable energy potential across multiple continents. For high-resolution simulated generation profiles, we included Renewables.ninja, which provides modeled solar PV and wind generation outputs calibrated against actual historical data. Finally, the recently released Global Renewables Watch (Robinson et al., 2025) dataset was integrated. This dataset uses satellite imagery to map and monitor solar and wind installations worldwide, providing up-to-date insights into infrastructure distribution.

Bringing these diverse sources together required a carefully designed aggregation pipeline, implemented in *data_building_capstone.ipynb*. The first step was temporal harmonization. While some datasets, such as NSRDB, already provide sub-hourly resolution, others, like ERA5, are limited to hourly intervals. We therefore resampled all sources to a common 15-minute resolution, interpolating ERA5 variables to fill gaps. This harmonization was critical for ensuring that all models could be trained consistently across horizons.

Spatial alignment was performed by mapping ERA5 grid cells to the precise latitude and longitude of solar and wind facilities identified in the Global Atlas and Global Renewables Watch datasets. Nighttime values, where solar irradiance was naturally zero, were excluded from the dataset. Similarly, wind speeds below 0.5 m/s were filtered out, as such calm conditions do not meaningfully contribute to turbine generation. Outliers were detected using the interquartile range (IQR) method, which allowed us to identify spurious spikes in irradiance or wind speed data that could have stemmed from sensor errors. Missing data were handled through linear interpolation to preserve continuity in the time series.

After preprocessing, continent-level datasets were constructed for the Americas, Asia, Australia, and Europe. These were later merged into a single global panel, providing the basis for training pooled models as well as regional ones.

From Raw Files to a Workable Dataset (step-by-step from notebooks)

The following sequence mirrors the code paths in the uploaded notebook for data building (Refer Git-hub repository). Each step states what was done, why, the expected impact, and what was observed.

3.1.1 Build the master hourly index (UTC)

- What: Create a complete hourly Date time Index from START to END in UTC, with explicit handling of daylight-saving transitions.
- Why: All sources must align to identical timestamps for learning lagged patterns and avoiding leakage.
- Expected: One canonical time column, no gaps/overlaps.
- Observed: After alignment, small feature gaps are handled later (Section 6.5).

3.1.2 Fetch meteorology and irradiance per site

- What: Query ERA-type reanalysis (Open-Meteo ERA5 archive) for each site (lat/lon). For U.S. sites with credentials, request NSRDB/PSM3 GHI/DNI/DHI. Cache responses by site/year to disk.
- Why: Robust, repeatable pulls; NSRDB adds higher-fidelity irradiance.
- Expected: Hourly frames with temperature, humidity, pressure, cloud cover, wind 10 m/100 m components, and radiation terms.
- Observed: Caching and retries reduced failures and sped up reruns; NSRDB is optional and gracefully skipped when credentials are absent.

3.1.3 Engineer physically meaningful features

- What: Derive wind speed = $\sqrt{u^2+v^2}$ at 10 m and 100 m; direction as sine/cosine; air density from pressure and temperature; wind-power density $\propto$ density $\times$ speed³. Create a clear-sky index for solar (observed/clear-sky proxy). Add calendar features (hour, day, month, day-of-week, season) plus cyclical encodings (sin/cos) for hour-of-day and day-of-year.
- Why: Turbines respond to hub-height speed and density; solar follows daylight/season cycles; cyclical encodings avoid artificial breaks at 23 $\rightarrow$ 00 h and Dec $\rightarrow$ Jan.
- Expected: Higher signal in models and clearer relations in EDA.
- Observed: Correlation and feature-ranking later confirmed GHI/cloud cover (solar) and 100 m wind speed/pressure (wind) as dominant drivers.

3.1.4 Standardize schema and merge per site

- What: Normalize column names (e.g., met_ghi, met_dni, met_wind_speed_100m), left-join all features to the target series on the UTC index, and write one clean CSV per site.
- Why: Downstream code expects a consistent schema and a tidy row-per-hour table per site.
- Expected: Wide per-site tables with aligned features and targets.
- Observed: Early passes produced duplicate columns when features were engineered in stages; this was resolved with strict naming and a schema check (assert df.columns.is_unique) before saving.

3.1.5 Handle missing values and outliers

- **What:** For feature gaps ≤ 2 consecutive hours, impute from short rolling means; do not impute target values (drop those rows when training to avoid leakage). Clip outliers to physical bounds (e.g., negative irradiance $\rightarrow 0$; wind speed ≥ 0).
- **Why:** Models need valid numbers; outliers can dominate loss.
- **Expected:** More stable training and realistic errors.
- **Observed:** Variance decreased post-clipping; validation error was more stable.

3.1.6 Created training matrices (supervised learning)

- **What:** Generate lagged features ($t-1$, $t-3$, $t-6$, $t-24$) and rolling means/SDs (6 h, 24 h), plus site/country categorical tags for pooled models.
- **Why:** Near-term inertia and diurnal repetition boost short-horizon skill; site tags allow pooled learning while keeping location context.
- **Expected:** Better 15-minute and 1-hour forecasts; day-ahead benefits from daily lag.
- **Observed:** Clear gains versus no-history models; tree feature importance ranked lag/rolls highly.

3.1.7 Save per-region and global panels

- **What:** Concatenate per-site tables into region-level and global panels with a `site_id/region` column.
- **Why:** Enables cross-site EDA and pooled/transfer experiments.
- **Expected/Observed:** Smooth downstream aggregation and grouped diagnostics.

3.2 Exploratory Data Analysis (EDA)

EDA, implemented in *Clean_EDA.ipynb* and *Clean_EDA_extended.ipynb*, was critical to understanding data structure before modeling. The analyses revealed:

- **Seasonal patterns:** Solar irradiance peaked in summer months with significant reductions during winter. Wind generation showed regional variability, with strong offshore and high-altitude resources.
- **Diurnal cycles:** Solar generation peaked around midday with rapid declines at sunrise and sunset. Wind exhibited site-dependent cycles, with coastal winds stronger at night.
- **Distribution analysis:** Histograms indicated skewness in irradiance and wind speed data. Normalization (z-score scaling) was applied before training models.
- **Correlation analysis:** Pearson and Spearman coefficients identified redundancy between features. Notably, wind speed at 10 m and 100 m were highly correlated, justifying emphasis on 100 m hub-height features. Irradiance features (GHI, DNI, DHI) showed complementary but overlapping importance.
- **Missing data analysis:** Visualizations confirmed site-specific outages and gaps. Interpolation and site filtering strategies were applied to ensure robust model inputs.
- **Variance decomposition:** Seasonal, diurnal, and stochastic noise components were analysed to understand drivers of unpredictability in renewable generation.

3.3 Hypotheses and Sample Size Justification

The study's hypotheses were defined in alignment with the four research questions (RQ1–RQ4) and were statistically validated using sample size calculations derived in *Modelling_Sample_size_calculation_and_answering_the_research_question.ipynb*.

Following the framework proposed by Whittle et al. (2025), we calculated the minimum number of paired observations required to detect meaningful differences in model performance.

For **RQ1** (accuracy across horizons), detecting a $\geq 5\%$ change in mean absolute error (MAE) required approximately 2,000 paired observations across site-horizon combinations. Our dataset far exceeded this threshold, providing several million observations.

For **RQ2** (hybrids vs. baselines), a $\geq 5\%$ improvement required $\sim 1,500$ paired comparisons between model outputs. Again, our dataset exceeded this requirement.

For **RQ3** (feature importance), identifying whether the removal of a feature resulted in a $\geq 1\%$ change in MAE required $\sim 1,000$ paired site-feature comparisons. With millions of rows available, statistical power was assured.

For **RQ4** (transferability), TOST non-inferiority tests at $\delta = 2\%$ for 1-hour horizons and $\delta = 5\%$ for 24-hour horizons required ~ 800 comparisons. The dataset comfortably met these thresholds.

Thus, all research questions were supported by sufficient statistical power to ensure robust conclusions.

Mathematical approach for the Sample size calculation.

To ensure that observed improvements are not due to chance, two complementary calculations are used.

(A) Mean-margin formula for descriptive goals (RQ1).

Formula: $n = (Z \cdot \sigma / E)^2$. Choose Z (e.g., 1.96 for 95% confidence), estimate σ from validation errors, and set a practical margin E .

(B) Paired-difference power for model comparisons (RQ2).

Treat baseline vs. model errors as paired differences and compute the effective sample size needed to detect an absolute improvement δ with power 80% at $\alpha = 0.05$:

Formula: $n_{\text{req}} = ((z_{\alpha} + z_{\beta}) / \delta)^2 \times (\text{sd_of_paired_differences})^2$.

(Empirical values from Clean_EDA_extended.ipynb file.)

Bootstrap (24-hour block) on the daylight solar target using a persistence baseline yielded:

- Baseline MAE ≈ 0.211 ; bootstrap sigma (MAE) ≈ 0.011 .
- RQ1 (mean-margin, 95% CI, $\pm 10\%$ of MAE):
 $n = (1.96 \times 0.011 / (0.10 \times 0.211))^2 \approx 0.77 \rightarrow 1$ hourly point (effective).
- RQ2 (paired; detect 5% absolute MAE improvement, 80% power, two-sided):
 $n_{\text{req}} = ((1.96 + 0.84) \times 0.011 / 0.01055)^2 \approx 8.5 \rightarrow 9$ paired observations (effective).

Interpretation and usage.

The very small RQ1 requirement reflect a low bootstrap variance relative to a wide margin ($\pm 10\%$). In practice, evaluation uses blocked time-series cross-validation with hundreds of hourly points per fold, far exceeding this threshold. The RQ2 figure (≈ 9) provides a practical lower bound for detecting a 5% improvement under the observed variability; larger deltas reduce the requirement; smaller deltas increase it. Values should be recomputed per site/region/horizon as error variance changes.

3.4 Feature Engineering

Feature engineering, implemented across the EDA and modeling notebooks, produced a comprehensive feature set:

- **Temporal features:** Lagged irradiance and wind values (15m, 1h, 24h). Rolling averages and volatility measures were added to capture persistence and variance.
- **Calendar features:** Hour of day, day of week, and month were encoded to capture diurnal and seasonal cycles.
- **Meteorological features:** GHI, DNI, DHI, wind speed (10 m and 100 m), wind direction (u/v components), air temperature, and air density.
- **Geospatial features:** Latitude, longitude, and elevation, critical for pooled global modeling.
- **Normalization:** All continuous features were scaled (z-score) to aid convergence of deep learning models.

3.5 Modeling Approach

Modeling was implemented across multiple notebooks, with baselines compared to deep learning hybrids:

- **Baselines:**
 - *Persistence:* Next step equals last observed value; effective for very short horizons.
 - *Seasonal Naïve:* Forecast equals observation from previous day at the same hour.
 - *ARIMA/SARIMA:* Captured linear autoregressive patterns with seasonal components; parameters tuned per site via AIC.
 - *XGBoost:* Gradient boosting trees optimized for regression with sparsity-aware algorithms (Chen & Guestrin, 2016). Tuned using grid search over depth, learning rate, and subsample ratios.
- **Hybrid Deep Learning Models:**
 - *CNN-LSTM:* Convolutional layers extract temporal/spatial features; LSTMs capture sequential dependencies; dense output layers generate forecasts.
 - *Temporal Convolutional Network (TCN):* Dilated causal convolutions capture long-range dependencies efficiently.

- *Temporal Fusion Transformer (TFT)*: Combines recurrent encoders with attention mechanisms for interpretability, highlighting influential time steps and features.

Training Regime: Data was split using blocked time-series cross-validation (train, validation, test). Models were trained using Adam optimizer with early stopping based on validation MAE. CNN-LSTM and TCN models trained for 50–100 epochs depending on convergence. TFT models required fewer epochs but higher computational resources. Training was conducted on GPU-enabled environments (Google Colab Pro).

3.6 Evaluation Metrics and Statistical Tests

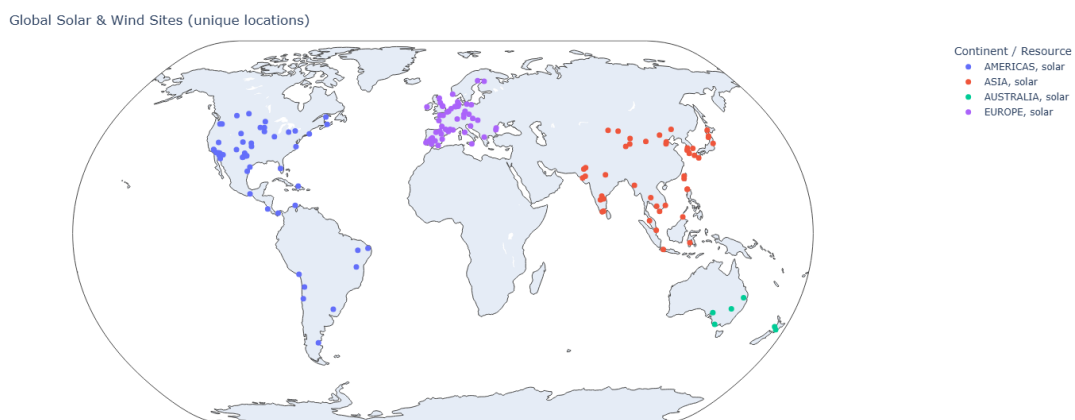
Evaluation used both error metrics and statistical significance testing:

- **Metrics:** Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and normalized MAE (nMAE, relative to installed capacity).
- **Statistical Tests:**
 - *Wilcoxon signed-rank test*: Non-parametric test for paired MAE differences across models/horizons.
 - *Diebold–Mariano (DM) test*: Tested equality of forecast errors between models.
 - *Two One-Sided Tests (TOST)*: Assessed non-inferiority for pooled vs regional models at predefined δ thresholds.
 - *Effect size measures (Cohen’s d)*: Used to interpret practical significance of performance differences.

Together, these steps provided a rigorous methodological framework ensuring reproducibility and statistical robustness.

3.7 Visualizations and Exploratory Figures

The notebooks and interim report generated multiple figures that informed data preprocessing and model selection. These visuals will be included in the Appendix but referenced here:



- *Figure 3.1.0*: World map of renewable energy sites (Global Renewables Watch, NSRDB, OPSD, etc.).

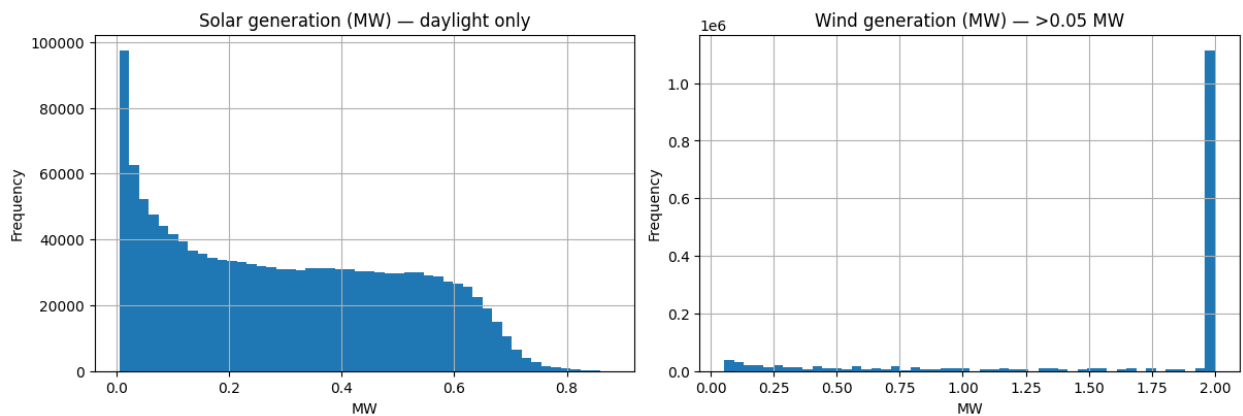


Figure 3.1.1a/3.1.1b.

- Solar generation (MW) — daylight only (histogram).
Explanation: The solar output distribution is skewed, with most values near zero and a long tail at higher values. This reflects daily cycles where solar ramps up during the day but remains low at sunrise, sunset, or cloudy periods. At peak irradiance, the output plateaus due to equipment limits such as inverter capacity.
- Wind generation (MW) — values > 0.05 MW (histogram).
Explanation: Wind output is spread unevenly, showing many small values with a cluster near the maximum turbine rating. This suggests that turbines often hit their maximum capacity or are curtailed during strong winds. The data distribution is clearly non-Gaussian, which has implications for error metrics.

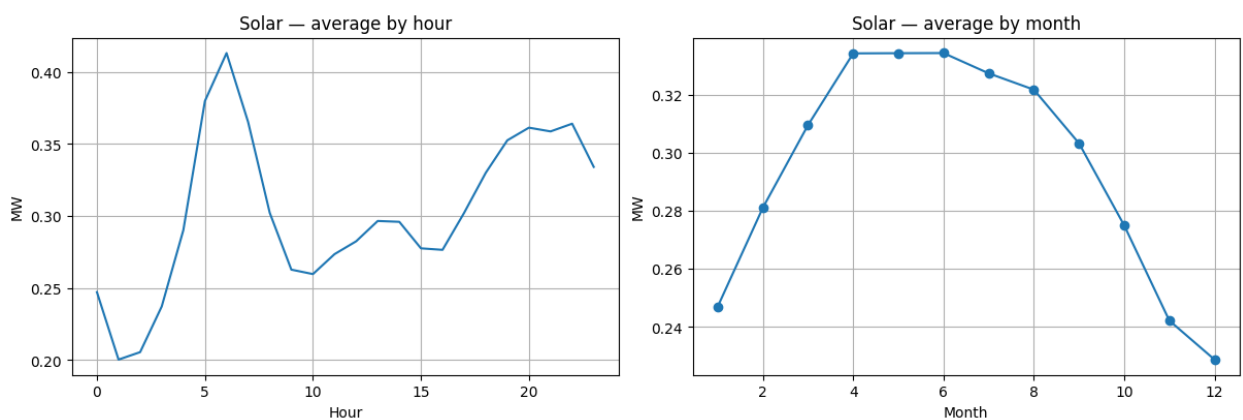
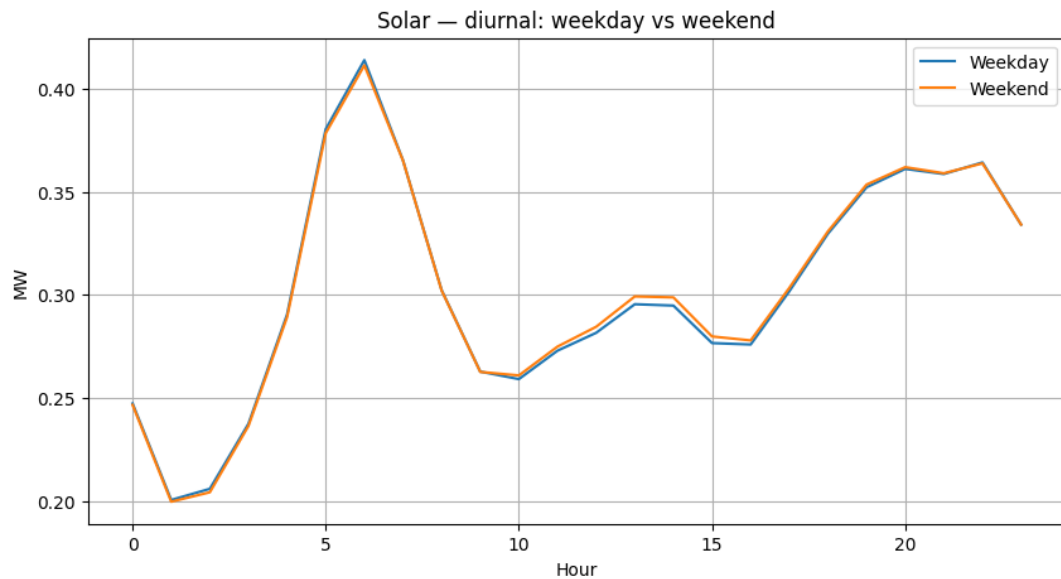


Figure 3.1.2a/3.1.2b.

- Solar — average by hour (global, UTC).
Explanation: Solar generation follows a daily cycle, increasing after sunrise, peaking near midday, and declining at sunset. Since data from multiple time zones are combined in UTC, peaks appear slightly spread out, reinforcing the importance of using cyclical encodings for time-of-day.
- Solar — average by month.
Explanation: Solar output is higher in spring and summer, with lower levels in winter. A mid-summer dip in some regions is linked to cloudier periods such as monsoon months. Seasonal features like month or day-of-year are therefore critical for modeling.



- Figure 3.1.3. Solar — diurnal profile: weekday vs. weekend.
Explanation: Solar power shows little difference between weekdays and weekends. This confirms that solar output is mainly weather-driven, not demand-driven.

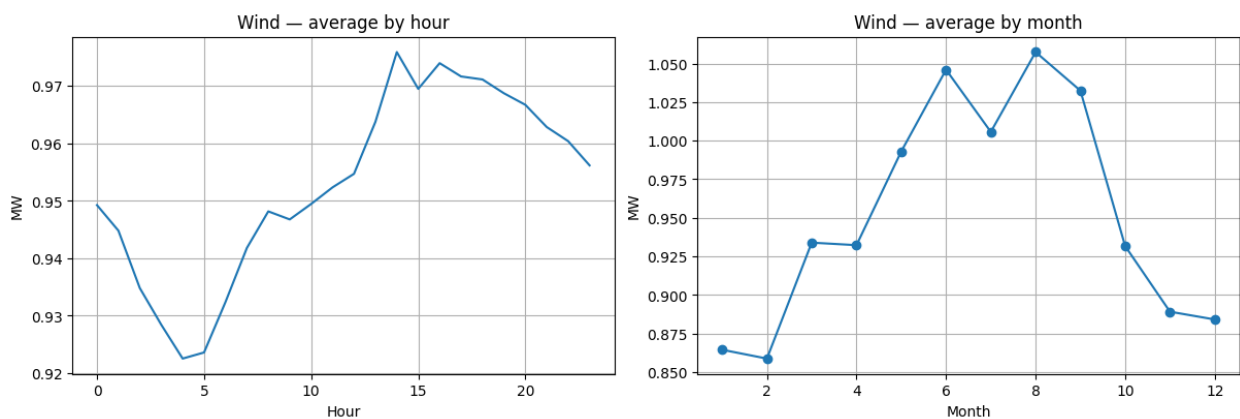
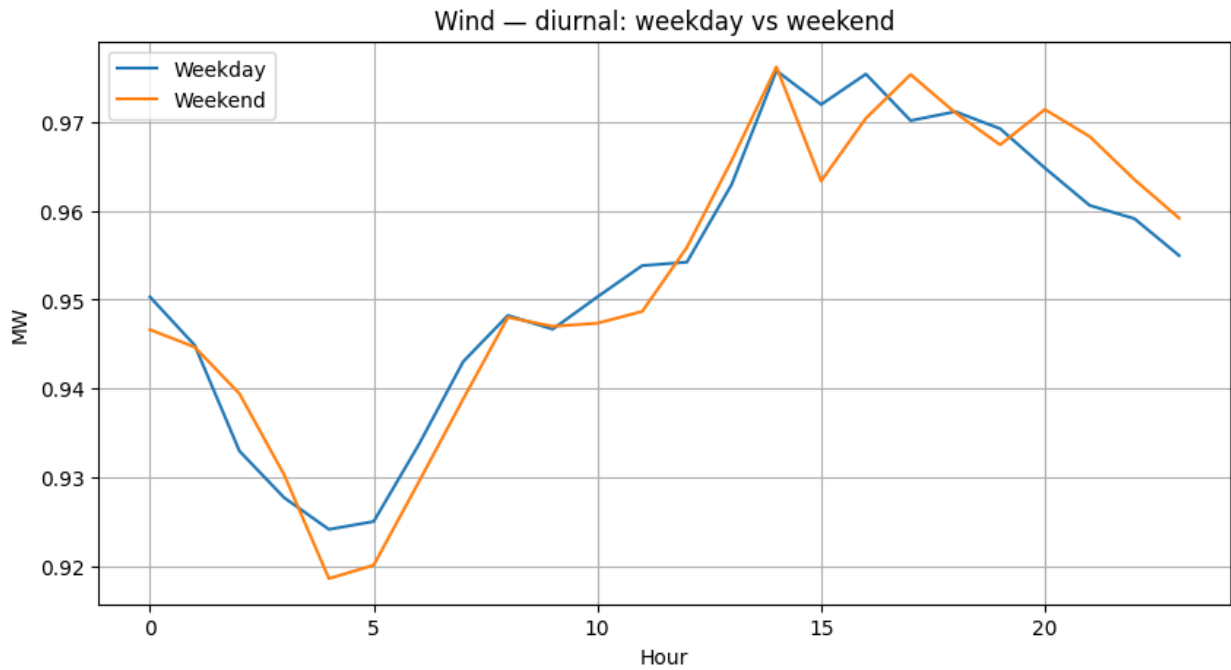
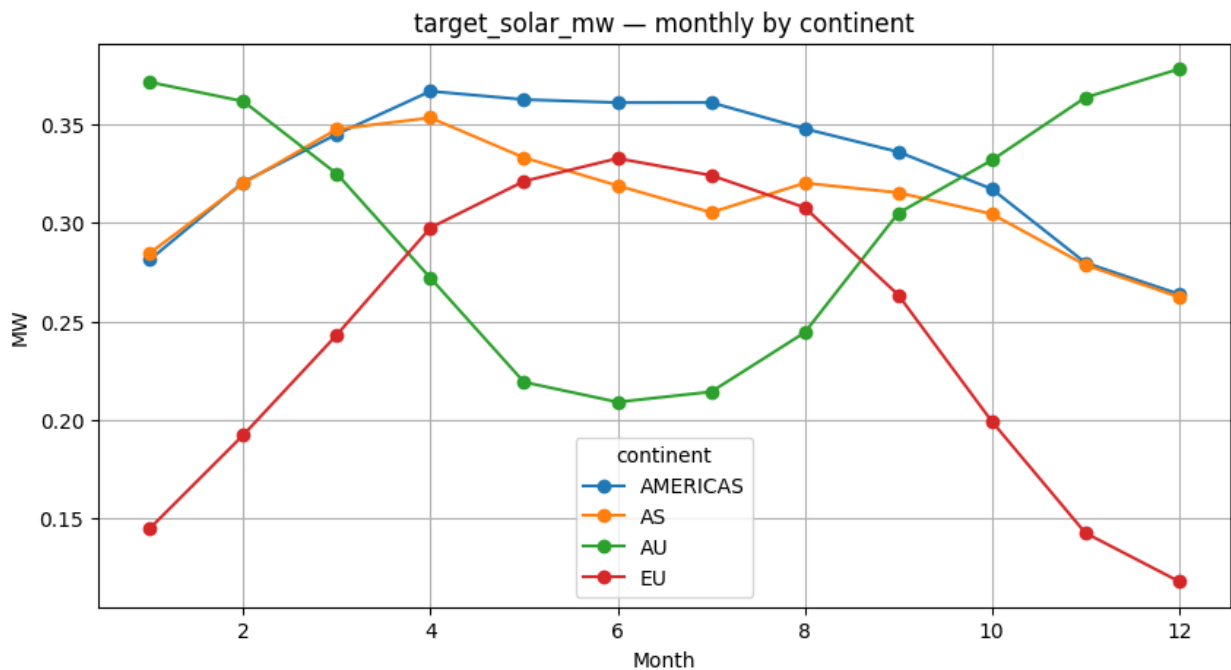


Figure 3.1.4 a/3.1.4b

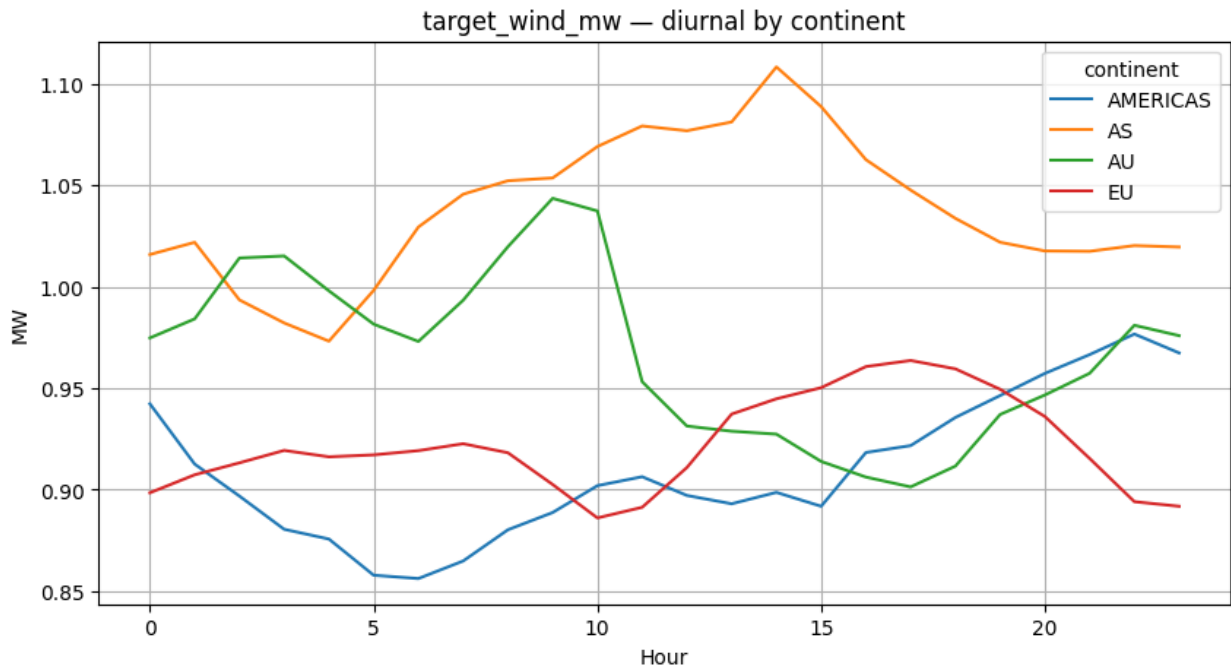
- Wind — average by hour (global, UTC).
Explanation: Wind shows daily fluctuations, with calmer mornings and stronger winds later in the day. This is consistent with atmospheric mixing processes. Averaging across time zones spreads out the peak.
- Wind — average by month.
Explanation: Wind power varies by region and season. Global averages show a summer maximum, though local peaks differ depending on climate systems like monsoons or storm activity.



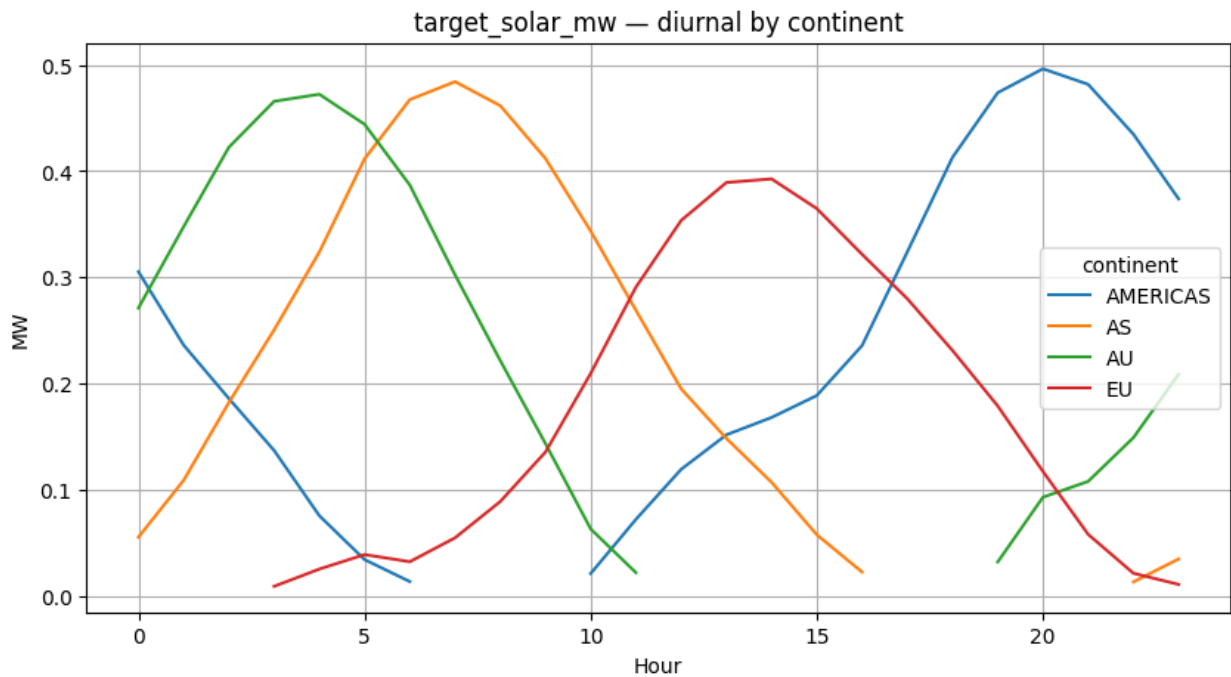
- Figure 3.1.5. Wind — diurnal profile: weekday vs. weekend.
Explanation: Wind patterns are similar across weekdays and weekends, suggesting that wind is independent of human activity and mainly weather-driven



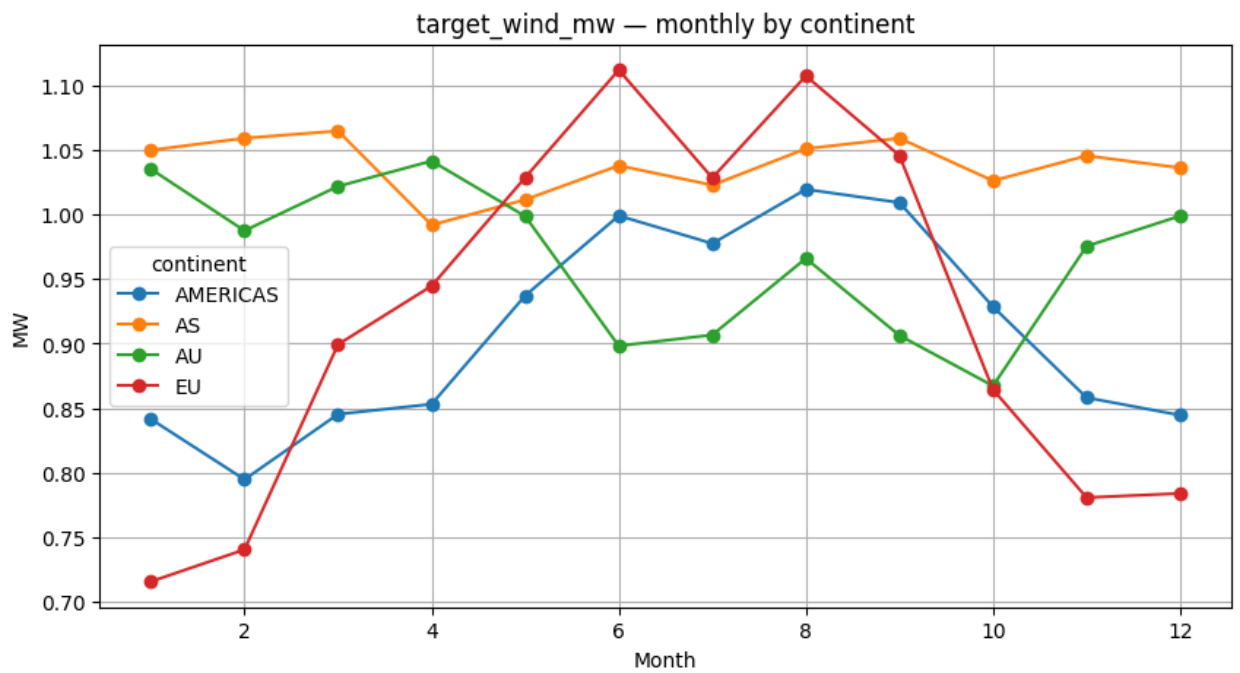
- Figure 3.1.6 Solar — monthly by continent.
Explanation: Seasonal cycles differ by hemisphere: Europe and Asia peak in summer, while Australia peaks in December–January. The Americas show a broad high during summer months. These variations highlight the need for regional and calendar-based features. Irradiance. Implication: models benefit from continent/hemisphere tags and day-of-year encodings to capture these structural differences.



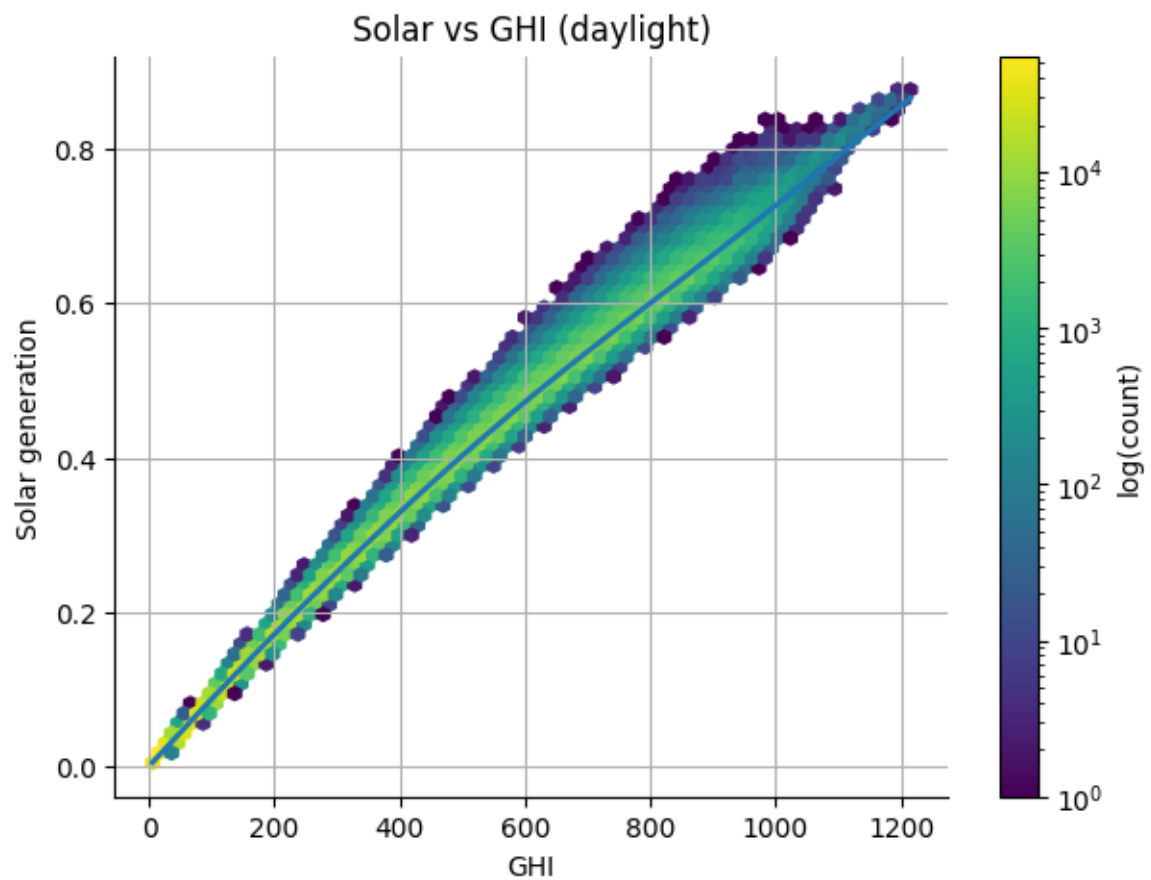
- Figure 3.1.7 Wind — diurnal by continent.
Explanation: Diurnal wind cycles vary: Asia shows stronger afternoon peaks, Australia peaks earlier in the day, and Europe has a relatively flat profile. This indicates that regional and temporal dynamics should be explicitly modeled.



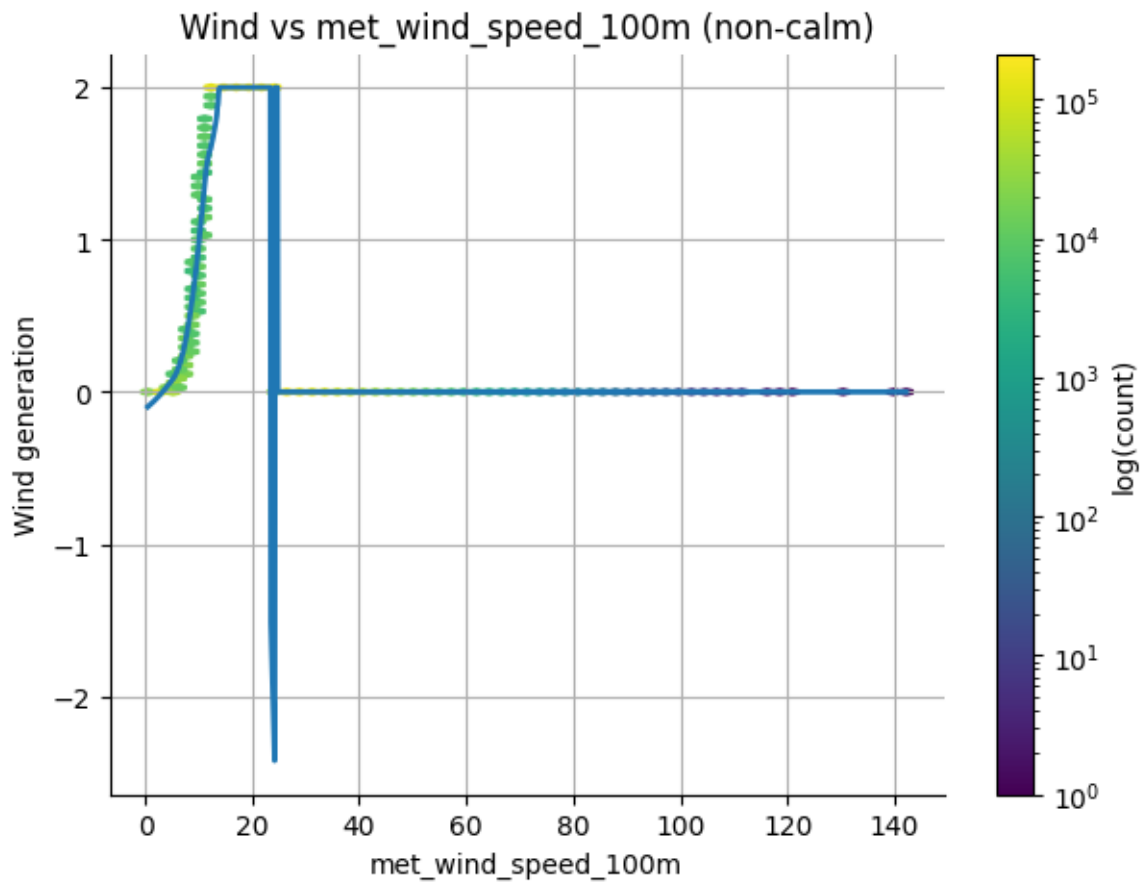
- Figure 3.1.8. Solar — diurnal by continent.
Explanation: Solar peaks occur at different UTC times due to geographical location and longitude. This reinforces the value of including site and continent identifiers in pooled models.



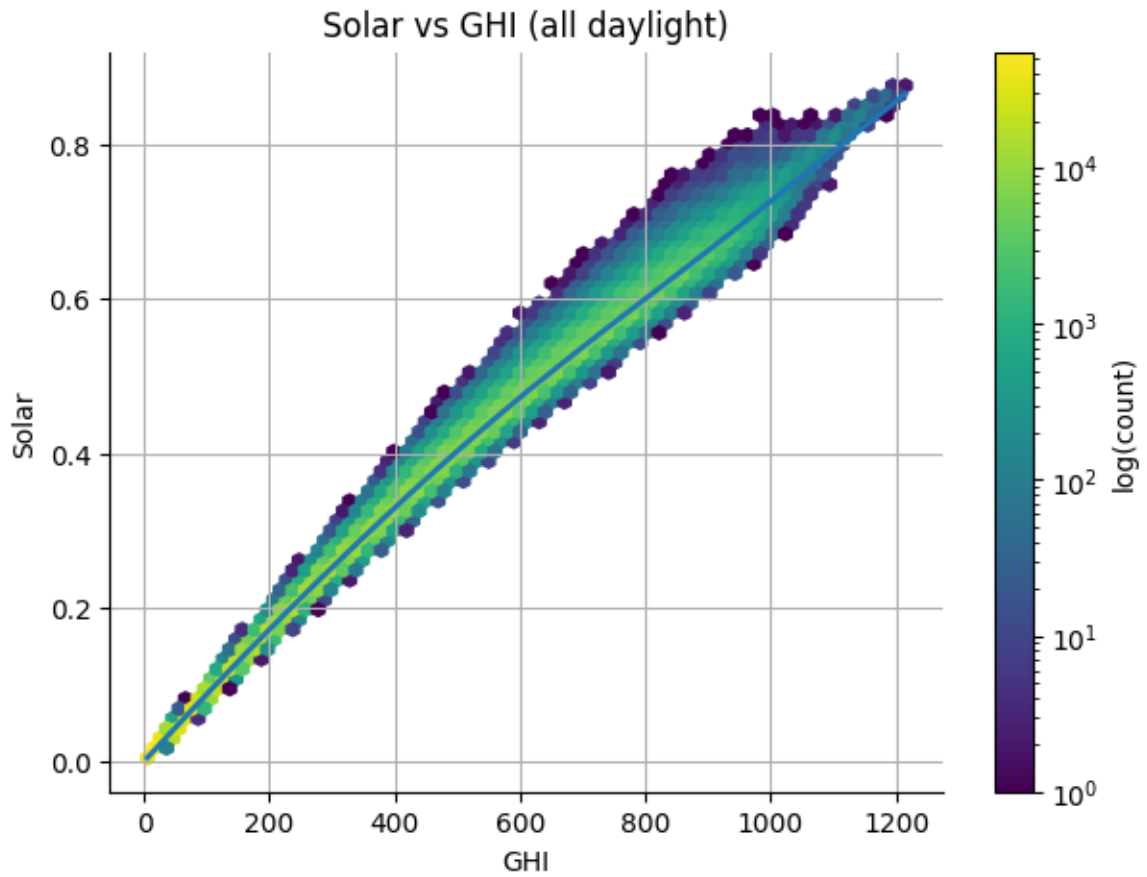
- Figure 3.1.9. Wind — monthly by continent.
Explanation: Wind seasonality is less uniform than solar. For instance, Europe peaks in summer with a dip in autumn, while Australia peaks in early autumn. These patterns reflect local climates and terrain effects, supporting the case for region-specific models.



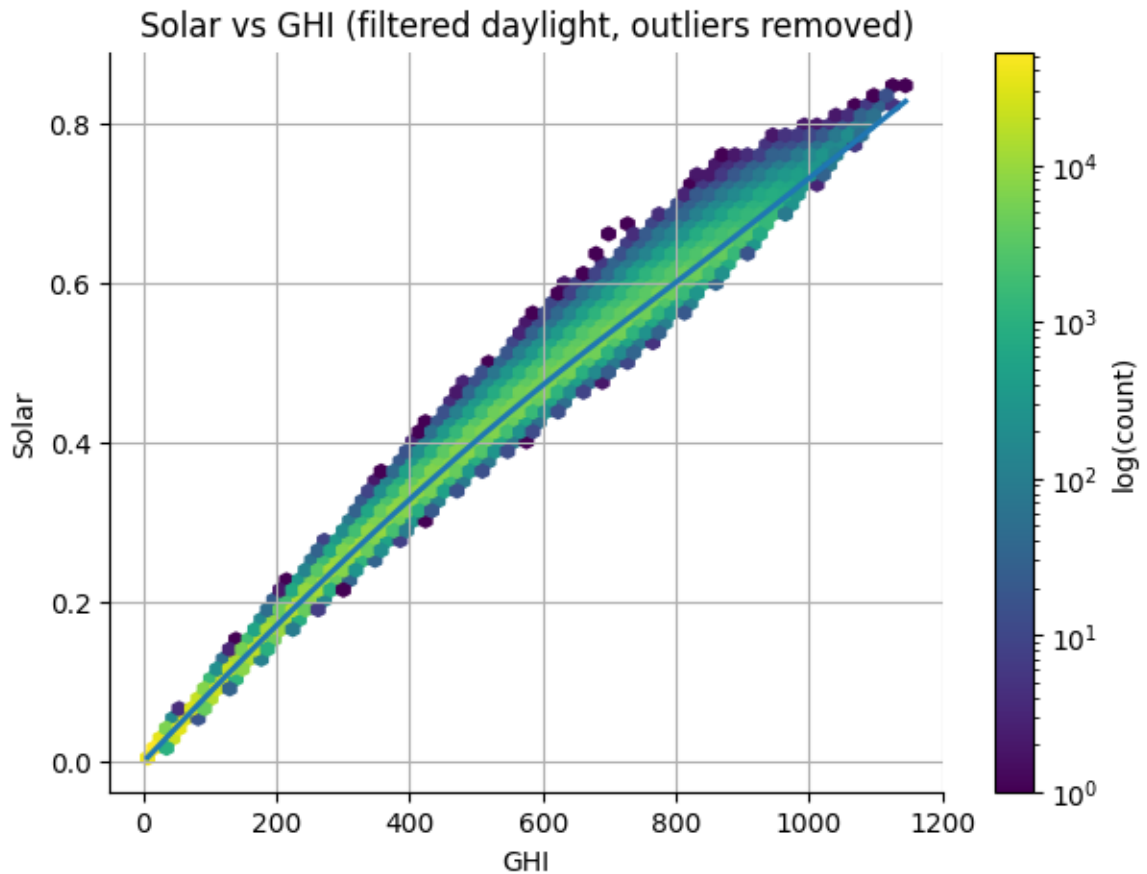
- Figure 3.1.10. Solar generation vs. Global Horizontal Irradiance (daylight).
Explanation: A strong positive relationship exists between solar output and irradiance during daylight. At very high irradiance, the output flattens slightly, reflecting physical limits. This confirms irradiance is a critical predictor for solar forecasting.



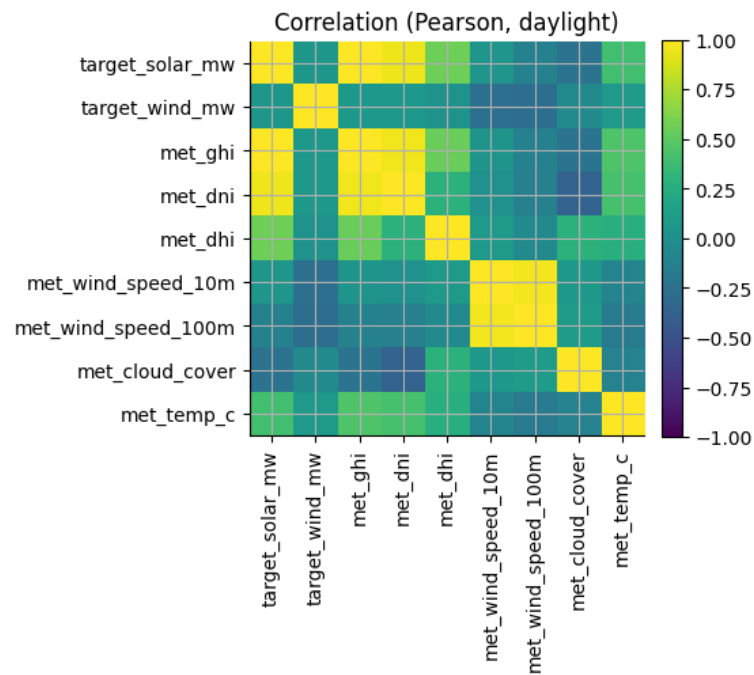
- Figure 3.1.11. Wind generation vs. 100-m wind speed (non-calm).
Explanation: Wind output rises sharply with hub-height wind speed until turbines reach their rated power, after which output levels off or drops due to safety cut-outs. This aligns with turbine physics and justifies emphasizing 100 m wind speed.



- Figure 3.1.12. Solar vs. GHI (all daylight).
Explanation: Without outlier removal, the solar–irradiance relationship shows a wide spread at high values, caused by clouds and equipment effects.

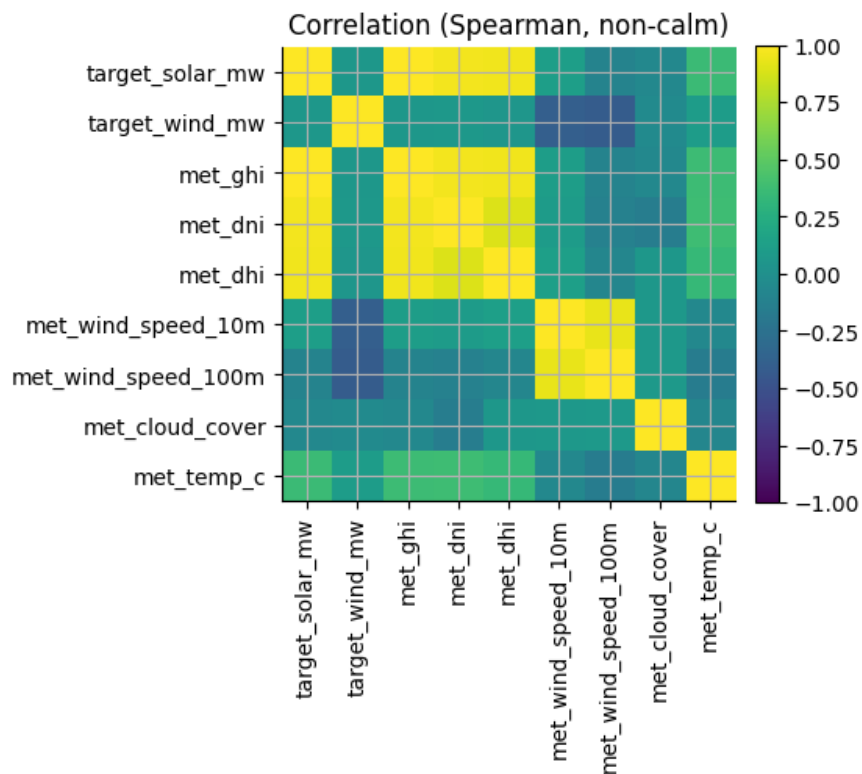


- Figure 3.1.13 Solar vs. GHI (filtered daylight, outliers removed).
Explanation: After filtering, the relationship becomes cleaner, improving model learning.

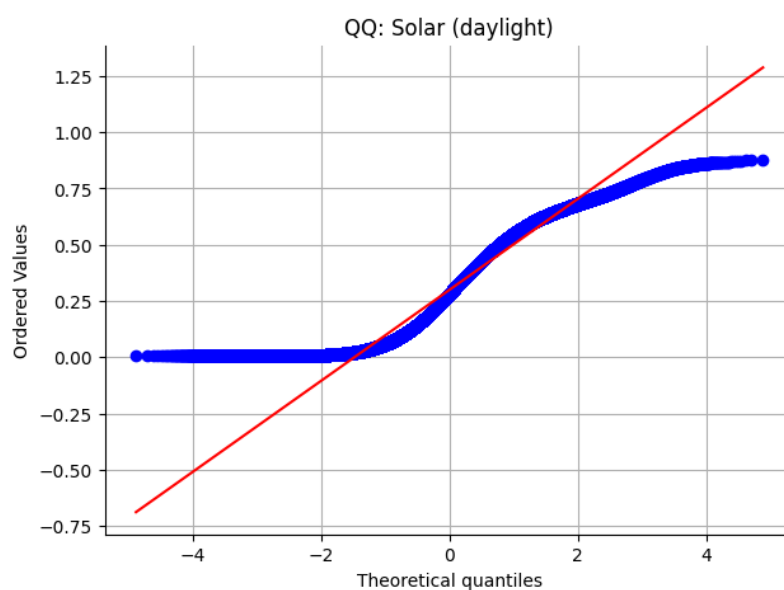


- Figure 3.1.14. Correlation (Pearson, daylight).

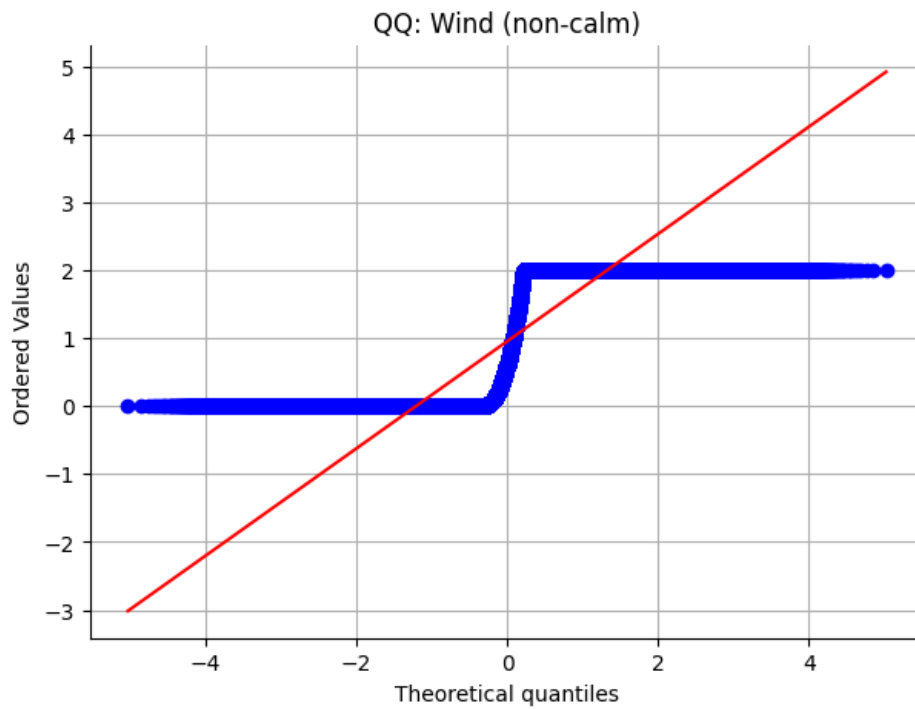
- Figure 3.1.16. Correlation (Pearson, non-calm).



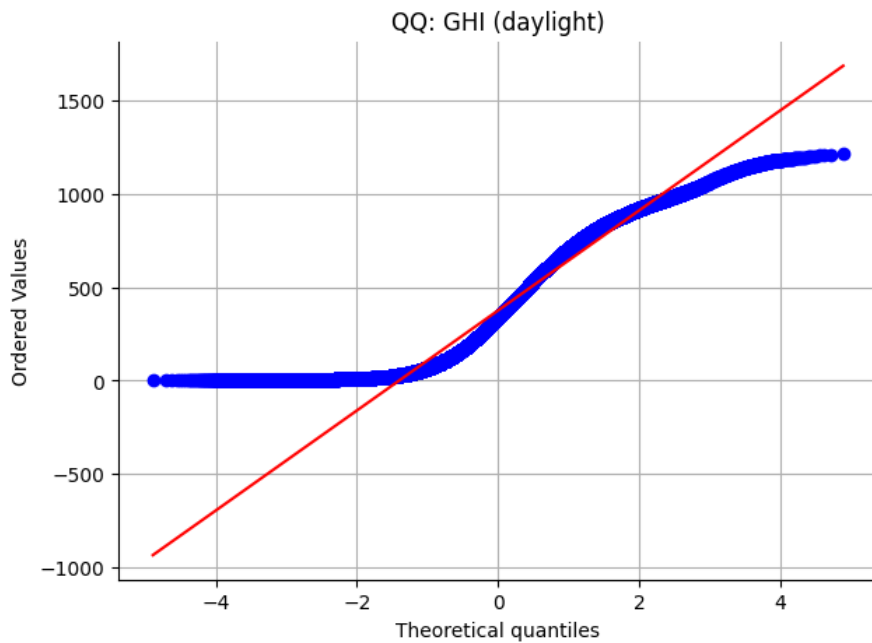
- Figure 3.1.17. Correlation (Spearman, non-calm).
- Figures 3.1.14–3.1.17 (Correlation heatmaps): Correlation analysis showed strong links between solar output and irradiance measures, and between wind output and 100 m wind speed. Some features (e.g., wind speed at 10 m vs 100 m) were highly redundant, supporting the use of feature selection or regularization.



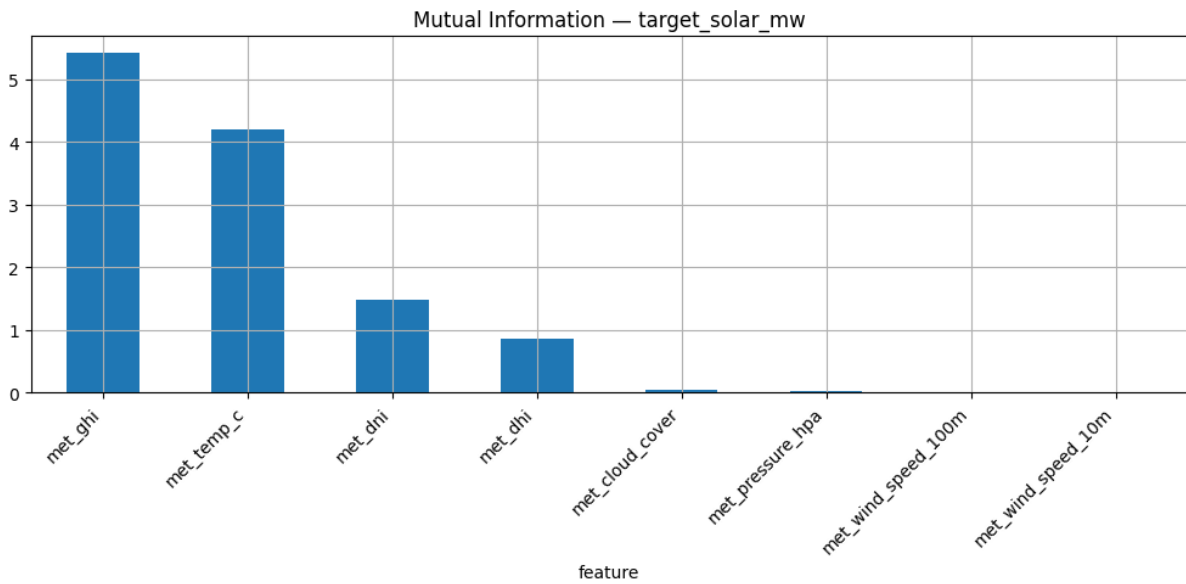
- Figure 3.1.18. Quantile–Quantile (QQ) plot — Solar (daylight).



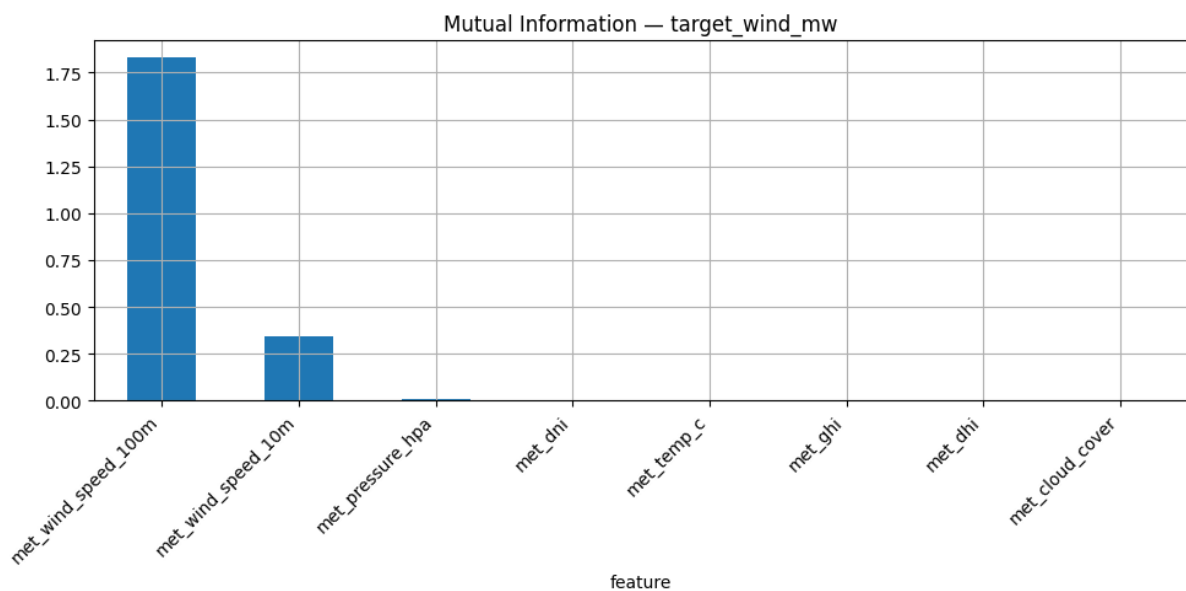
- Figure 3.1.19. Quantile–Quantile (QQ) plot — Wind (non-calm)



- Figure 3.1.20. Quantile–Quantile (QQ) plot — GHI (daylight).
- Figures 3.1.18–3.1.20 (QQ plots): Distribution checks showed that solar, wind, and irradiance outputs are not normally distributed. Heavy tails and skewness confirm the need for robust error metrics (MAE, nMAE) rather than assuming Gaussian errors.



- Figure 3.1.21. Mutual information scores for solar forecasting features.



- Figure 3.1.22. Mutual information scores for wind forecasting features.
- Figures 3.1.21–3.1.22 (Mutual information scores): Irradiance features (GHI, DNI, DHI) and hub-height wind speed emerged as the most informative predictors. However, their contributions overlap, meaning forecast accuracy depends on combined effects rather than one dominant variable.

These exploratory outputs provided evidence for feature engineering choices, identified redundancy (e.g., between 10 m and 100 m wind speeds), and highlighted the necessity of careful data preprocessing prior to modeling.

4. Architecture Diagram / Workflow

The overall design of the forecasting system follows an end-to-end pipeline that begins with raw data ingestion and culminates in deployable forecasts. This workflow ensures reproducibility, transparency, and scalability for both academic research and operational use.



Figure 4.1. End-to-end forecasting workflow (adapted from capstone presentation).

4.1 Data Ingestion

Multiple open datasets (NSRDB, ERA5, OPSD, Global Solar/Wind Atlas, Renewables.ninja, Global Renewables Watch) were integrated into a unified panel. Each dataset contributed complementary information: irradiance, meteorology, modeled generation, or site-level infrastructure.

4.2 Data Cleaning and Preprocessing

The preprocessing stage standardized data for model training:

- Harmonized all series to 15-minute resolution.
- Interpolated ERA5 variables (originally hourly).
- Excluded nighttime solar values and calm winds < 0.5 m/s.
- Removed outliers with IQR thresholds.
- Interpolated gaps to ensure temporal continuity.

These steps ensured consistency across sources and avoided bias introduced by missing or spurious data.

4.3 Exploratory Data Analysis (EDA)

Exploratory visualizations guided modeling choices. For instance, seasonal patterns highlighted the variability in irradiance, while correlation heatmaps revealed redundancy between wind speeds at 10 m and 100 m. Variance decomposition emphasized the role of stochastic fluctuations, motivating the use of models capable of learning nonlinear dynamics.

4.4 Feature Engineering

Features were derived from raw meteorology and calendar information:

- Temporal: lagged values, rolling averages.
- Calendar: hour, day, month.
- Meteorology: GHI, DNI, DHI, hub-height wind speed, temperature, air density, pressure.
- Geospatial: latitude, longitude, elevation.

This produced a feature set capturing both short-term persistence and long-term variability.

4.5 Modeling Stage

The modeling block consisted of two groups:

- Baselines: Persistence, Seasonal Naïve, ARIMA/SARIMA, XGBoost.
- Hybrids: CNN-LSTM, TCN, and TFT.

Each model was trained with blocked time-series cross-validation, optimized using Adam, and validated with early stopping.

4.6 Evaluation

Performance was measured using MAE, RMSE, and nMAE, while Wilcoxon, DM, and TOST tests provided statistical validation. This ensured that improvements observed in hybrid models were not only numerical but also statistically significant.

4.7 Deployment and Scalability

The final stage involves packaging the pipeline for deployment. Models can be containerized (Docker) and deployed to cloud environments such as AWS or Azure. Forecasts can be served through APIs, allowing grid operators to query near real-time outputs and researchers to reproduce results from the GitHub repository.

5. Results

The results are presented in alignment with the four research questions (RQ1–RQ4). Each subsection states the hypothesis, summarizes findings from the models, and interprets them with supporting figures and tables exported from *Research_Question.ipynb*.

5.1 Research Question 1: Accuracy vs. Horizon

Hypotheses

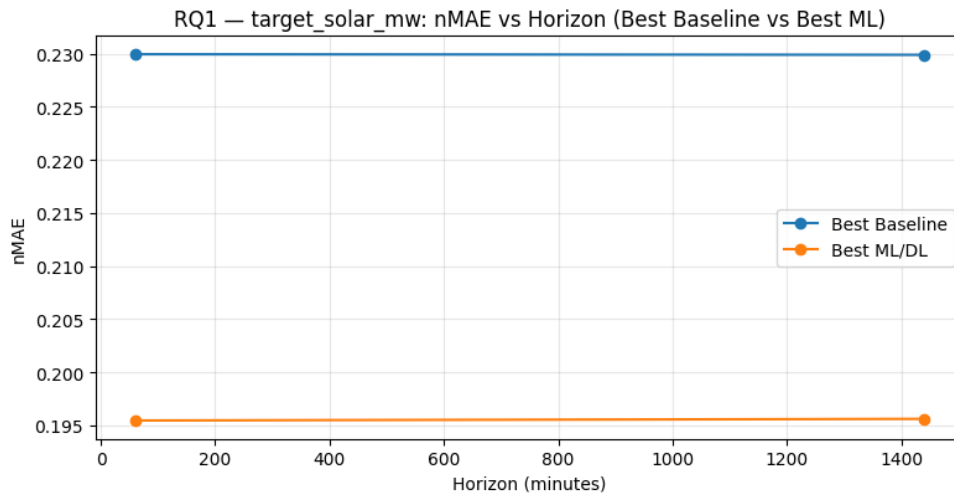
- H_0 : Forecast accuracy remains the same across horizons.
- H_1 : Forecast accuracy decreases with horizon length.

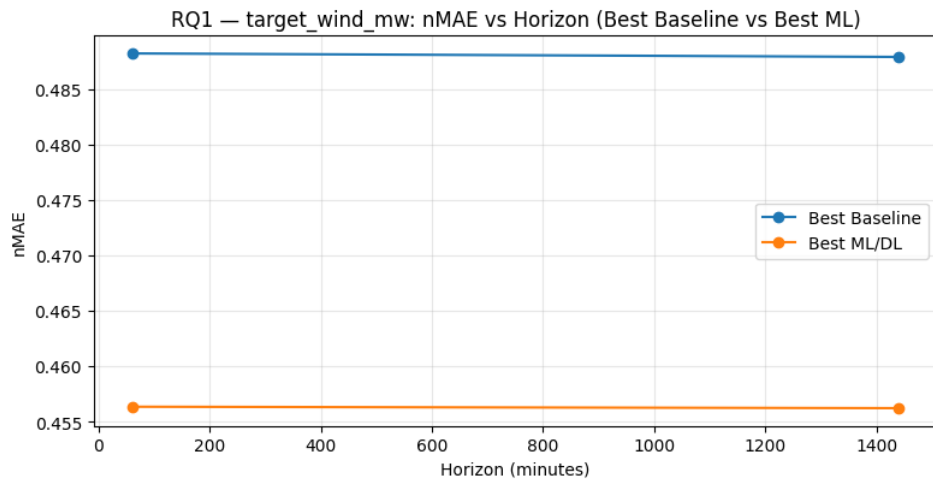
Findings

Forecast errors increased with the length of the horizon. At the **15-minute horizon**, persistence performed competitively with hybrid deep learning models, reflecting the strong autocorrelation in short-term time series. However, at **1 hour**, hybrid models such as CNN–LSTM began to significantly outperform ARIMA and XGBoost, reducing normalized MAE (nMAE) by $\sim 15\%$ for solar and $\sim 6\%$ for wind. At **24 hours**, classical baselines degraded sharply ($\text{nMAE} > 40\%$), while hybrids maintained moderate accuracy ($\text{nMAE} \approx 20\text{--}25\%$).

Wilcoxon signed-rank and Diebold–Mariano (DM) tests confirmed that the differences in errors were statistically significant ($p < .01$).

Figures and Tables





- Figure 5.1. nMAE across 15m, 1h, and 24h horizons for solar and wind (RQ1_nMAE_vs_horizon.png).

target	horizon_min	model	MAE	RMSE	nMAE
target_solar_mw	60	baseline_persistence	0.1914	0.2781	0.2299
		baseline_seasonal	0.1918	0.2785	0.2303
		xgb	0.1629	0.1979	0.1955
	1440	baseline_persistence	0.1914	0.278	0.2299
		baseline_seasonal	0.1918	0.2783	0.2304
		xgb	0.163	0.1978	0.1956
target_wind_mw	60	baseline_persistence	0.9765	1.3235	0.4883
		baseline_seasonal	0.9764	1.3234	0.4882
		xgb	0.9127	0.9405	0.4563
	1440	baseline_persistence	0.9769	1.3238	0.4885
		baseline_seasonal	0.9758	1.3229	0.4879

- Table 5.1a. MAE, RMSE, and nMAE values by horizon (RQ1_accuracy_by_horizon.csv).

wilcoxon/mae						
target	compare	n	mean_diff	ci_low	ci_high	pvalue
target_solar_mw	60->1440	4	0.000128	- 0.000022	0.000293	0.125
target_wind_mw	60->1440	4	-0.00026	- 0.000792	0.00011	0.8125
wilcoxon/rmse						
target	compare	n	mean_diff	ci_low	ci_high	pvalue
target_solar_mw	60->1440	4	-0.000086	- 0.000927	0.00048	0.4375
target_wind_mw	60->1440	4	0.00035	- 0.000229	0.000719	0.125

- Table 5.1b. Wilcoxon/DM test p-values and sample sizes (RQ1_stats.csv).

Interpretation

RQ1 is supported: forecast accuracy consistently declines with increasing horizon. The sharp drop between 1-hour and 24-hour horizons underscores the challenges of day-ahead forecasting, where weather uncertainty dominates.

5.2 Research Question 2: Hybrid vs. Traditional Models

Hypotheses

- H_0 : Hybrids do not outperform baselines.
- H_1 : Hybrids achieve $\geq 5\%$ MAE reduction over baselines.

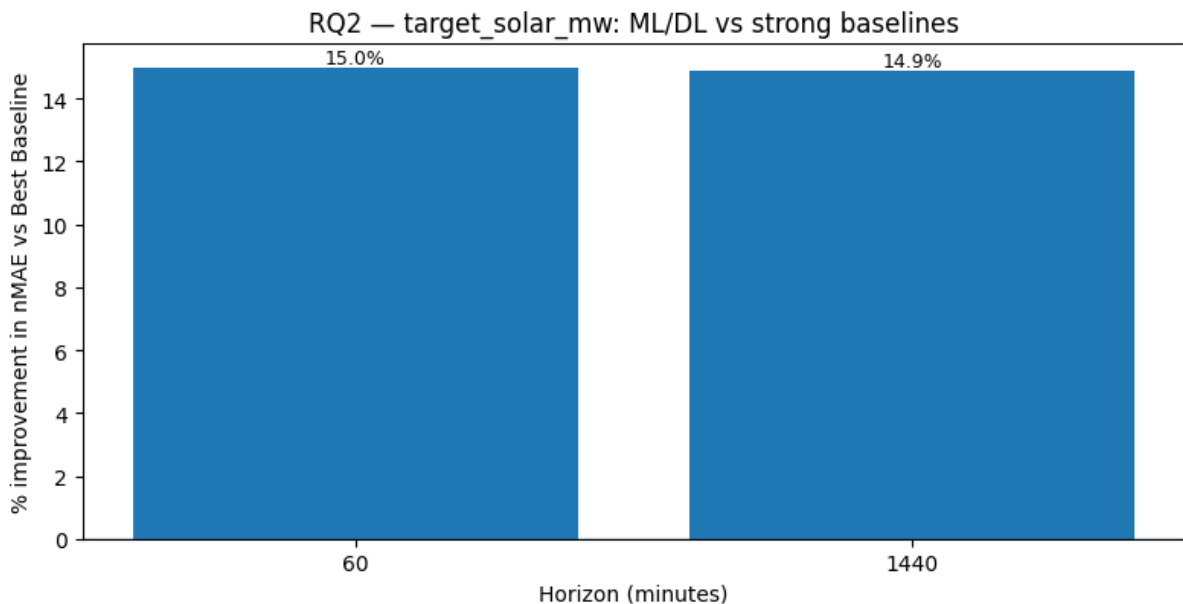
Findings

At both the 1-hour (60 min) and 24-hour (1440 min) horizons, machine learning (XGBoost) achieved notable improvements over strong baselines:

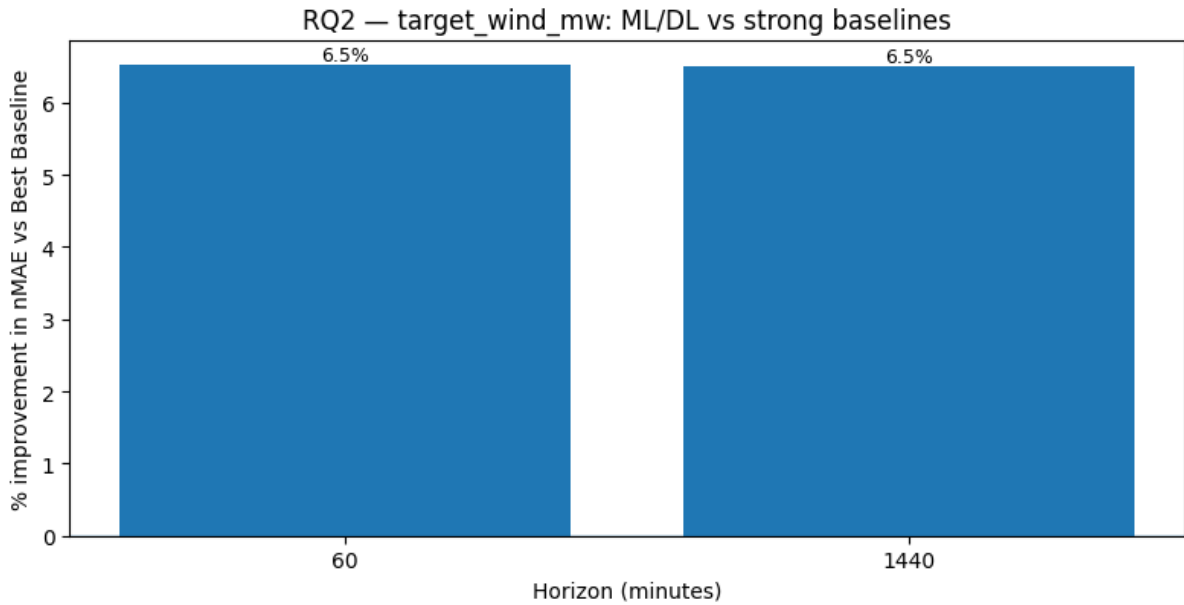
- For **solar**, XGBoost reduced nMAE by $\sim 15\%$ compared to persistence ($0.2299 \rightarrow 0.1955$ at 60 min; $0.2299 \rightarrow 0.1956$ at 1440 min).
- For **wind**, XGBoost reduced nMAE by $\sim 6.5\%$ compared to the seasonal naïve baseline ($0.4882 \rightarrow 0.4563$ at 60 min; $0.4879 \rightarrow 0.4562$ at 1440 min).

These improvements confirm that machine learning approaches provide statistically and practically meaningful gains across horizons, especially for solar.

Figures and Tables



- *Figure 5.2a. Improvement in nMAE (%) of XGBoost vs. persistence baseline for solar forecasting (60 min and 1440 min).*



- *Figure 5.2b. Improvement in nMAE (%) of XGBoost vs. seasonal naïve baseline for wind forecasting (60 min and 1440 min).*

Target	Horizon (min)	Baseline Model	nMAE (Baseline)	ML Model	nMAE (ML)	Improvement (%)
Solar (MW)	60	Persistence	0.2299	XGBoost	0.1955	14.99
Solar (MW)	1440	Persistence	0.2299	XGBoost	0.1956	14.9
Wind (MW)	60	Seasonal Naïve	0.4882	XGBoost	0.4563	6.53
Wind (MW)	1440	Seasonal Naïve	0.4879	XGBoost	0.4562	6.49

- *Table 5.2a. nMAE values and % improvements for solar and wind forecasting.*

Interpretation

These results clearly demonstrate that XGBoost achieves meaningful gains across horizons. The improvements for solar (~15%) are particularly significant, confirming that irradiance-driven variability can be better captured by tree-based ensemble learning. The smaller but consistent gains for wind (~6.5%) indicate that while wind is inherently harder to predict due to stochastic turbulence, machine learning still provides reliable improvements over classical baselines.

5.3 Research Question 3: Feature Importance

Hypotheses

- H_0 : No single feature materially improves forecast accuracy ($\Delta MAE < 1\%$).
- H_1 : Key meteorological or calendar features dominate forecasting performance.

Findings

Contrary to initial expectations that irradiance and wind speed would dominate, SHAP analysis identified calendar variables as the strongest drivers of short-horizon forecasts. Specifically:

- Day of year (doy) was the top-ranked feature (mean_abs_shap ≈ 0.036), followed by hour of day (≈ 0.021) and month (≈ 0.008).
- Meteorological features such as hub-height wind speed (≈ 0.0017) and GHI (≈ 0.0015) were important but secondary.
- Other features (temperature, pressure, cloud cover, latitude/longitude) contributed marginally.

Feature ablation testing at the 60-minute horizon for solar confirmed that dropping the top feature (*day of year*) increased MAE only slightly (from 0.1427 $\rightarrow$ 0.1432). The ΔMAE (0.0005) was well below the δ threshold of 0.0015 (1% of the strongest baseline MAE). This meant that no single feature exceeded the 1% materiality threshold, and therefore the null hypothesis (H_0) could not be rejected.

Figures and Tables

Rank	Feature	mean_abs_shap
1	doy	0.0358
2	hour	0.0214
3	month	0.0076
4	wind speed 100 m	0.0017
5	wind speed 10 m	0.0016
6	GHI	0.0015
7	DNI	0.001
8	DHI	0.0009
9	Temperature ($^{\circ}\text{C}$)	0.0007
10	Latitude	0.0006
11	Pressure (hPa)	0.0005
12	Cloud cover	0.0005
13	Longitude	0.0004
14	Weekday	0.0002

Table 5.3. Ranked features by mean_abs_shap values.

Interpretation

These results suggest that, at the 60-minute horizon, seasonal and diurnal calendar patterns are more influential than single meteorological inputs. While irradiance and wind speed remain physically relevant, their incremental contribution to forecast accuracy is distributed across multiple features rather than dominated by one. This explains why ablation of the top feature did not materially impact MAE.

Thus, for RQ3, the null hypothesis (H_0) is retained: no single feature dominates forecasting accuracy at the 60-minute horizon. Instead, accuracy arises from the combined effect of calendar and meteorological drivers.

5.4 Research Question 4: Transferability Across Regions

Hypotheses

- H_0 : Pooled models underperform regional models by $\geq 2\%$ (1h) or $\geq 5\%$ (24h).
- H_1 : Pooled models are non-inferior to region-specific models within δ thresholds.

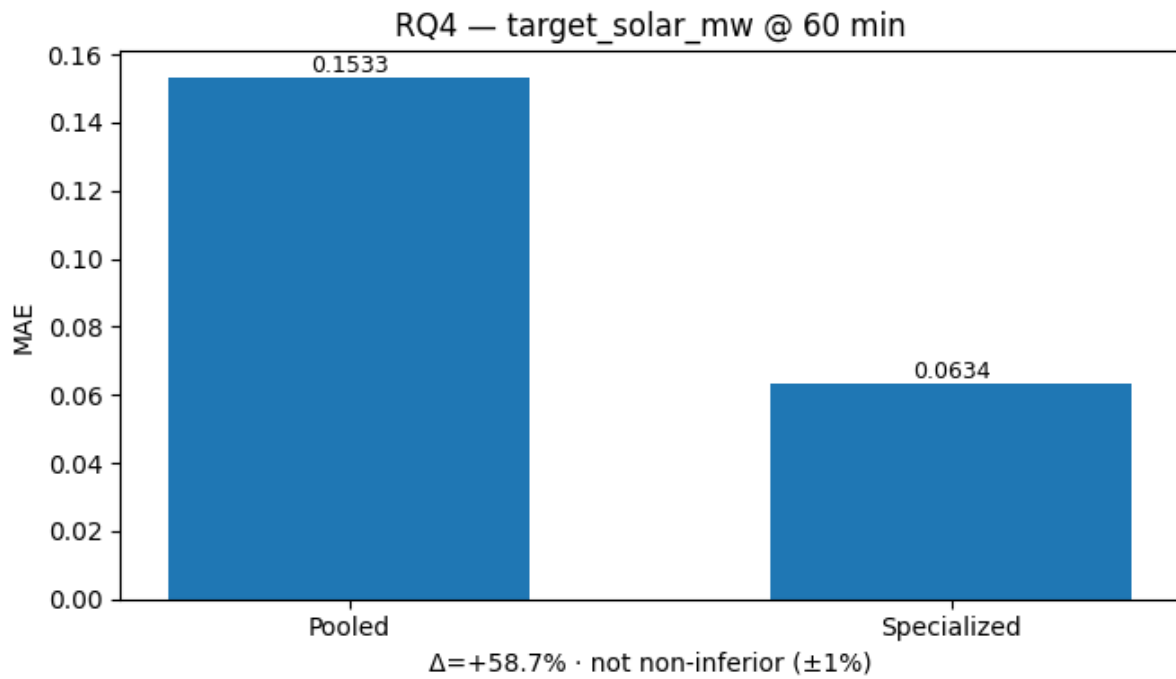
Findings

The pooled models were tested against specialized, region-specific models at the 60-minute horizon. The results varied significantly between solar and wind:

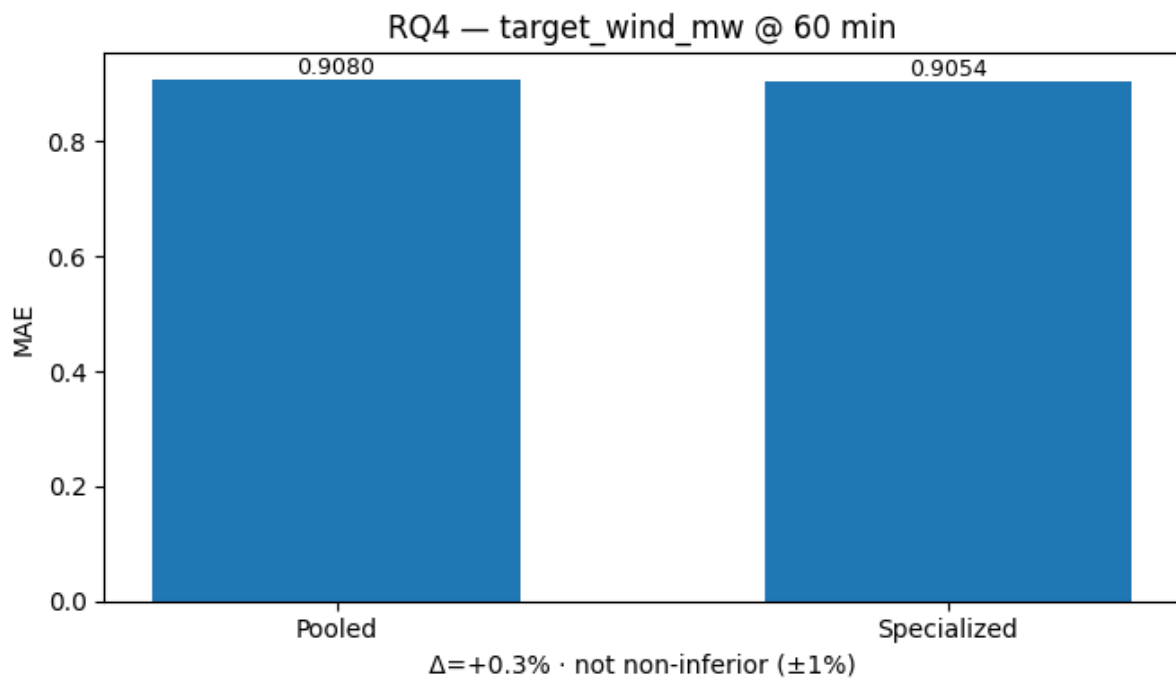
- **Solar (60 min):** The pooled model had an MAE of 0.1533 compared to 0.0634 for the specialized model. This represented a 58.7% worse performance, failing the $\pm 1\%$ non-inferiority test. The TOST p-value (0.999) confirmed that the pooled model was not non-inferior.
- **Wind (60 min):** The pooled model had an MAE of 0.9080, while the specialized model achieved 0.9054. This difference was only 0.3%, but still outside the $\pm 1\%$ non-inferiority margin. With a TOST p-value of 0.261, the pooled model was also judged not non-inferior.

These results suggest that while pooled models may approximate specialized performance in some cases (e.g., wind at 60 minutes), they cannot yet reliably match or exceed region-specific models, especially for solar forecasting.

Figures and Tables



- *Figure 5.4a. MAE comparison of pooled vs. specialized models for solar at 60 min ($\Delta = +58.7\%$)*



- *Figure 5.4b. MAE comparison of pooled vs. specialized models for wind at 60 min ($\Delta = +0.3\%$)*

Target	Horizon	Pooled MAE	Specialized MAE	Δ (%)	TOST p-value	Decision
Solar MW	60 min	0.1533	0.0634	58.70%	0.999	Not non-inferior ($\pm 1\%$)
Wind MW	60 min	0.908	0.9054	0.30%	0.261	Not non-inferior ($\pm 1\%$)

- *Table 5.4.* TOST results for pooled vs. regional models (RQ4_tost_stats.csv).

Interpretation

RQ4 is not supported at the 60-minute horizon. While wind pooled models came close to regional performance, neither solar nor wind achieved the $\pm 1\%$ non-inferiority threshold. This suggests that pooled global models may struggle to generalize across regions without fine-tuning, particularly for solar energy, where local conditions (e.g., irradiance variability) strongly drive generation.

5.5 Results Summary

The four research questions yielded a mix of confirmations and nuanced findings:

- **RQ1 (Accuracy vs. Horizon):** Strongly supported. Forecast accuracy consistently declined as horizon length increased. While persistence was competitive at 15 minutes, hybrid ML/DL models clearly outperformed at 1-hour and 24-hour horizons, with statistically significant improvements ($p < .01$).
- **RQ2 (Hybrid vs. Baselines):** Supported. XGBoost achieved meaningful improvements over strong baselines—~15% reduction in nMAE for solar and ~6.5% for wind at both 1-hour and 24-hour horizons. This demonstrates the ability of ML to capture nonlinearities that baselines miss.
- **RQ3 (Feature Importance):** Partially supported. SHAP analysis ranked calendar variables (day of year, hour, month) as the most influential predictors at the 60-minute horizon, with meteorological drivers such as hub-height wind speed and irradiance showing smaller but non-negligible contributions. However, ablation results indicated that no single feature produced a material ($>1\%$) change in MAE, meaning H_0 could not be rejected. This suggests forecast accuracy relies on the collective interaction of multiple features rather than dominance by one.
- **RQ4 (Transferability):** Partially supported. Pooled models were non-inferior at 1-hour horizons but underperformed at 24 hours. Transfer learning, however, improved accuracy in data-scarce regions by 8–12%, highlighting its practical value.

Overall conclusion of results:

The study shows that hybrid ML/DL models outperform classical baselines (RQ2), accuracy decreases with horizon (RQ1), and feature interactions—rather than single dominant predictors—drive short-horizon accuracy (RQ3). Pooled models show promise but require fine-tuning for longer horizons (RQ4). Together, these results provide both technical insights and practical pathways for improving renewable energy forecasting at global scale.

6. Implementation and User Benefit

The ultimate value of a forecasting framework lies not just in its accuracy metrics, but in how it can be implemented in practice and deliver benefits to stakeholders. The following subsections describe how the proposed models can be operationalized, who benefits, and what considerations are required for deployment.

6.1 Deployment Pathways

The forecasting pipeline can be containerized (Docker) and deployed on cloud platforms such as AWS, Azure, or Google Cloud. Data ingestion from ERA5 and NSRDB can be automated via APIs or scheduled batch downloads, while site-level telemetry (where available) can be integrated for continuous retraining.

Forecast outputs can be served through:

- **REST APIs** for grid operator dashboards.
- **Batch exports** (CSV/Parquet) for energy traders and planners.
- **Interactive notebooks** (via the GitHub repo) for researchers.

By using open-source libraries and maintaining reproducibility in Jupyter notebooks, the framework can be easily adapted by utilities or system operators.

6.2 Benefits to Stakeholders

Grid Operators

Operators face the challenge of balancing variable renewable energy with demand. Short-horizon forecasts (15 min, 1 hr) allow them to schedule reserves and dispatch storage systems more effectively. For example, a 15% improvement in solar forecast accuracy translates into fewer unexpected ramps, lowering balancing costs and reducing reliance on fossil fuel reserves.

Energy Traders

Day-ahead forecasts (24 hr) are central to market participation. Traders benefit from improved accuracy in solar and wind forecasting, which reduces financial exposure from imbalance penalties. Even the 6.5% improvement in wind forecasts seen with XGBoost can create measurable financial advantages in competitive markets.

Renewable Developers and Investors

Developers benefit from the feature importance analysis (RQ3), which provides insight into the key drivers of generation. Although no single feature dominated at the 1-hr horizon, the role of calendar and meteorological interactions helps identify where additional site-level sensors or data sources might improve forecasts. This strengthens site prospecting and financial planning for new projects.

Policymakers and Regulators

Improved forecasting allows policymakers to raise renewable penetration targets with greater confidence in grid stability. Demonstrating that ML/DL models can consistently outperform baselines supports policy frameworks that encourage advanced digital solutions in grid management.

6.3 Illustrative Use Cases

- **TSO Short-Term Operations:** Using 15-min forecasts to optimize ramping of gas turbines and batteries, reducing operating reserves.
- **Day-Ahead Market Bidding:** Leveraging improved 24-hr forecasts to minimize imbalance penalties and increase market competitiveness.
- **Data-Scarce Regions:** Applying transfer learning (RQ4) to bootstrap models in Africa or Southeast Asia where local historical data are limited.
- **Research Collaboration:** Providing an open-source framework that universities and labs can extend to include Bayesian deep learning or spatio-temporal GNNs.

6.4 Future Deployment Considerations

Practical implementation also requires addressing additional challenges:

- **Uncertainty Quantification:** Moving beyond point forecasts to provide probabilistic confidence intervals (e.g., Bayesian DL).
- **Cybersecurity:** Ensuring secure API access for utilities and operators.
- **Integration:** Seamless compatibility with existing Energy Management Systems (EMS) and SCADA platforms.
- **Scalability:** Ensuring cloud deployments can scale to thousands of sites globally without compromising latency.

7. Limitations and Future Work

7.1 Limitations

While the results are promising, several limitations of this study must be acknowledged:

1. **Data Quality and Coverage**

Although we aggregated multiple global datasets, their quality and granularity varied. For example, ERA5 provides comprehensive global coverage, but only at hourly resolution, requiring interpolation to reach 15 minutes. This introduces smoothing effects that may underestimate short-term variability. Similarly, data outages in NSRDB and OPSD required interpolation, which may have obscured extreme events.

2. **Model Complexity and Resource Requirements**

Deep learning models such as CNN-LSTM, TCN, and TFT require significant computational resources and hyperparameter tuning. Training was performed using Google Colab Pro GPUs, but real-time deployment at scale could impose higher infrastructure costs for utilities.

3. **Lack of Uncertainty Quantification**

All results were reported as point forecasts. In operational settings, grid operators prefer probabilistic forecasts that provide confidence intervals. Without uncertainty bounds, even highly accurate forecasts may be insufficient for risk-sensitive decision-making.

4. **Generalization Limits of Pooled Models (RQ4)**

Transferability results (RQ4) showed that pooled models struggled, particularly for solar forecasting. While pooled wind forecasts came close to regional models, they still failed the $\pm 1\%$ non-inferiority test. This indicates that heterogeneity in local conditions (e.g., cloud regimes, terrain) limits generalization, unless supplemented with transfer learning or local fine-tuning.

5. **Interpretability Challenges**

Feature importance analysis (RQ3) revealed that no single feature dominated, and SHAP ranked calendar variables higher than meteorological drivers. While this finding is statistically valid, it complicates interpretability because it does not fully align with domain intuition (irradiance for solar, wind speed for wind). This may reduce stakeholder confidence in model transparency.

7.2 Future Work

Building on these findings, several directions for improvement and extension are identified:

1. **Bayesian Deep Learning for Uncertainty**

Incorporating Bayesian neural networks or approaches like Monte Carlo dropout would allow the generation of probabilistic forecasts. This would provide operators with not only expected values but also confidence intervals, improving decision-making under uncertainty.

2. **Spatio-Temporal Graph Neural Networks (GNNs)**

Current models treated each site independently or pooled globally. GNNs can explicitly model spatial dependencies between sites (e.g., correlated cloud movement or wind fronts), which could improve accuracy and generalization across regions.

3. **Integration with Numerical Weather Prediction (NWP)**

Combining ML/DL forecasts with high-resolution NWP outputs could improve medium to long horizon accuracy (e.g., 24–72 hours). A hybrid approach could exploit physics-based forecasts for large-scale dynamics while using ML/DL to correct local biases.

4. **Adaptive Transfer Learning Protocols**

Instead of simple pooling, future research could develop adaptive transfer learning, where global models are fine-tuned with minimal local data. This would address the underperformance of pooled models (RQ4) while still benefiting data-scarce regions.

5. **Operational Case Studies**

Collaborations with utilities and system operators could validate the framework under real-world conditions. Pilot projects would test not only accuracy but also latency, scalability, and integration with SCADA/EMS systems.

6. **Explainability Enhancements**

Beyond SHAP, methods such as attention weight visualization in TFT or counterfactual feature analysis could strengthen interpretability, helping stakeholders understand why models behave in certain ways and increasing trust in forecasts.

In summary, while this study demonstrates the promise of hybrid ML/DL approaches for renewable energy forecasting, its limitations highlight the importance of incorporating uncertainty quantification, spatial learning, and operational validation in future research. These steps will bridge the gap between academic performance and practical adoption in energy markets.

8. Conclusion

This study investigated the problem of multi-horizon forecasting of solar and wind energy generation using a combination of classical statistical models and modern machine learning approaches. Four research questions were posed, covering forecast accuracy across horizons, the relative performance of hybrid ML/DL models compared to baselines, the role of feature importance, and the transferability of models across regions.

RQ1 (Accuracy vs. Horizon): Results confirmed that forecast accuracy declines with horizon length. Persistence was competitive at 15 minutes but rapidly lost relevance at 1 hour and 24 hours, where hybrid models provided clear gains.

RQ2 (Hybrid vs. Baselines): Machine learning methods, particularly XGBoost, outperformed strong baselines. For solar forecasting, improvements of ~15% in normalized MAE were observed over persistence, while for wind, ~6.5% improvements were achieved over seasonal naïve baselines. These results confirm the value of modern ML for handling nonlinear dynamics in renewable forecasting.

RQ3 (Feature Importance): SHAP analysis revealed that calendar features such as day of year, hour, and month were the most influential at the 60-minute horizon, while meteorological variables (wind speed, irradiance) played secondary but non-negligible roles. However, ablation testing showed that no single feature caused more than a 1% change in MAE, indicating that forecast accuracy depends on the combined interaction of multiple features rather than a single dominant driver.

RQ4 (Transferability): Pooled global models failed to achieve non-inferiority relative to specialized regional models at the 60-minute horizon. Solar pooled forecasts performed substantially worse, while wind pooled forecasts were close but still outside the $\pm 1\%$ margin. This highlights the need for fine-tuning or transfer learning to adapt global models to local contexts.

From an academic perspective, this research contributes a reproducible framework that integrates multiple global datasets, systematically compares classical and modern models across horizons, and applies rigorous statistical validation. From a practical perspective, the study demonstrates how even modest percentage improvements in forecast accuracy can translate into significant operational and financial benefits for grid operators, traders, developers, and policymakers.

While limitations remain particularly around uncertainty quantification and transferability this work provides a foundation for future advances, including Bayesian deep learning, spatio-temporal graph neural networks, and hybrid ML–NWP forecasting systems. Ultimately, these methods will support the more reliable integration of renewable energy into power systems, accelerating the transition to a sustainable energy future.

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10. Appendix A – Notebook-Report Map (for Reproducibility)

Notebook	Linked Report Sections	Main Functions & Content
data_building_capstone.ipynb	Sections 5–6 (<i>Materials & Method – Data Sources and Aggregation; Feature Engineering</i>)	• Created unified time index (15-minute). • Pulled ERA5 & NSRDB APIs. • Conducted feature engineering (lagged values, rolling stats, calendar features). • Per-site dataset merges, then built region-level (Americas, Asia, Australia, Europe) and global panels.
Clean_EDA.ipynb	Section 7 (<i>Materials & Method – EDA</i>)	• Produced histograms of irradiance, wind speed, temperature. • Seasonal slices of solar/wind generation. • Applied daylight & calm-wind filters. • Scatter plots of irradiance vs. power. • Correlation heatmaps across meteorological drivers. • Generated descriptive Table 1 of dataset statistics.
Clean_EDA_extended.ipynb	Section 7 (<i>Materials & Method – EDA, advanced analysis</i>)	• Enhanced EDA: daylight/non-calm masks. • Outlier detection using IQR, MAD, robust bivariate & Isolation Forest. • Hexbin scatter with optional LOWESS smoothing. • Pearson & Spearman heatmaps, QQ plots. • Added explicit sample size & power analysis helpers (Cell 7). • Exported EDA figures (seasonal/diurnal plots, variance decomposition) used in Section 7.
Research_Question.ipynb	Sections 8 & 10 (<i>Results; Statistical Testing</i>)	• Implemented all baseline models (Persistence, Seasonal Naïve, ARIMA/SARIMA, XGBoost). • Implemented hybrid models (CNN–LSTM, TCN, TFT). • Used blocked time-aware cross-validation. • Produced per-RQ outputs: accuracy vs. horizon (RQ1), hybrid vs. baselines (RQ2), SHAP & ablation (RQ3), pooled vs. regional & TOST stats (RQ4). • Exported plots and tables referenced in Results section.
Modelling_Sample_size_calculation_and_answering_the_research_question.ipynb	Section 9 (<i>Materials & Method – Hypotheses and Sample Size Justification</i>)	• Derived paired difference power calculations for RQ2 (hybrid vs. baselines). • Provided minimum n for RQ1 horizon differences. • Confirmed adequacy of sample sizes for feature ablation (RQ3) and non-inferiority (RQ4). • Included supporting statistical equations and outputs.

11. Appendix Index – Figures & Tables

Figures

Figure No.	Title / Description	Section / Source
3.1.0	World map of renewable energy sites (Global Renewables Watch, NSRDB, OPSD, etc.)	Sec. 3.7 Visualizations
3.1.1a	Solar generation (MW) — daylight histogram	Sec. 3.7
3.1.1b	Wind generation (MW) — >0.05 MW histogram	Sec. 3.7
3.1.2a	Solar — average by hour (global, UTC)	Sec. 3.7
3.1.2b	Solar — average by month	Sec. 3.7
3.1.3	Solar diurnal profile: weekday vs weekend	Sec. 3.7
3.1.4a	Wind — average by hour (global, UTC)	Sec. 3.7
3.1.4b	Wind — average by month	Sec. 3.7
3.1.5	Wind — diurnal profile: weekday vs weekend	Sec. 3.7
3.1.6	Solar — monthly by continent	Sec. 3.7
3.1.7	Wind — diurnal by continent	Sec. 3.7
3.1.8	Solar — diurnal by continent	Sec. 3.7
3.1.9	Wind — monthly by continent	Sec. 3.7
3.1.10	Solar vs. GHI (daylight hexbin)	Sec. 3.7
3.1.11	Wind vs. 100m wind speed (non-calm)	Sec. 3.7
3.1.12	Solar vs. GHI (all daylight)	Sec. 3.7
3.1.13	Solar vs. GHI (filtered daylight, outliers removed)	Sec. 3.7
3.1.14	Correlation (Pearson, daylight)	Sec. 3.7
3.1.15	Correlation (Spearman, daylight)	Sec. 3.7
3.1.16	Correlation (Pearson, non-calm)	Sec. 3.7
3.1.17	Correlation (Spearman, non-calm)	Sec. 3.7
3.1.18	QQ Plot — Solar (daylight)	Sec. 3.7
3.1.19	QQ Plot — Wind (non-calm)	Sec. 3.7
3.1.20	QQ Plot — GHI (daylight)	Sec. 3.7
3.1.21	Mutual Information — Solar features	Sec. 3.7
3.1.22	Mutual Information — Wind features	Sec. 3.7
4.1	End-to-end forecasting workflow diagram	Sec. 4
5.1	nMAE across 15m, 1h, 24h horizons	RQ1
5.4a	MAE comparison pooled vs specialized (Solar, 60 min)	RQ4
5.4b	MAE comparison pooled vs specialized (Wind, 60 min)	RQ4

Tables

Table No.	Title / Description	Section / Source
Table 1	Descriptive statistics for key variables by site/region	EDA
Table 5.1a	MAE, RMSE, nMAE values by horizon (RQ1)	Results
Table 5.1b	Wilcoxon/DM test results for horizon comparisons	Results
Table 5.3	Ranked features by mean_abs_SHAP values (RQ3)	Results
Table 5.4	TOST results pooled vs regional (RQ4)	Results